A PROJECT REPORT ON

SMART WALKING STICK USING ESP32 AND ULTRASONIC SENSORS

Submitted in partial fulfillment of requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Submitted by:

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Under the Esteemed Guidance of
Dr. G. Chakrapani M. Tech., Ph.D.

Assistant Professor, Dept. of CSE (AI & ML)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY
(AUTONOMOUS)
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RECOGNIZED UGC-DDU KAUSHAL KENDRA
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YEAR: 2025-2026

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

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We have not submitted the matter embodied in this report for the award of any other Degree in any other institutions for the academic year 2025-2026.

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Whatever one does, whatever one achieves, the first credit goes to the **Parents** be it not for their love and affection, nothing would have been responsible. We see in every good that happens to us their love and blessings.

BY

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ABSTRACT

The Smart Walking Stick is a tool that helps blind people get around more easily and be more independent. Blind people have to rely on other people to help them find their way when they walk in the traditional way, which makes them less confident and free. To fix this problem, the proposed system uses modern embedded technology, like an ESP32, and multiple sensors to find obstacles and environmental conditions in real time. The system has four ultrasonic sensors: one on the front, one on the left, one on the right, and one on the bottom. These sensors help the system find things like pits or manholes that could be dangerous. The buzzer makes different levels of alerts, like medium, high, and emergency tones, to warn the user based on the distance that was found.

The system has a soil moisture sensor that can tell if there is water on the ground, which makes navigation safer. A mobile app made with MIT App Inventor gives voice feedback to let users know exactly where obstacles are, which helps them be more aware of their surroundings. Also, a GPS module is built in to make things safer by sending an emergency message with the user's location to a caretaker when needed. The suggested system is cost-effective, efficient, and greatly lowers the need for help from others, which makes life better for blind people.

Keywords: Walking stick, ESP32, ultrasonic sensors, soil moisture sensor, GPS module, emergency button, and buzzer

CONTENTS

Chapter No		Title	Page No
		List of Abbreviations	IV
		List of Tables	V
		List of Figures	VI
1.		**INTRODUCTION TO SMART WALKING STICK**	
1.1		Overview of Smart Walking Stick	1
1.2		Need for the Project	2
1.3		Objectives of the Project	3
1.4		Applications of Smart Walking Stick	4
1.5		Advantages of the System	5
1.6		Summary	6
2.		**LITERATURE REVIEW**	
2.1		Introduction to survey	7
2.2		Related Works	8
	2.2.1	Related Work – 1	8
	2.2.2	Related Work – 2	9
	2.2.3	Related Work – 3	10
	2.2.4	Related Work – 4	11
	2.2.5	Related Work – 5	12
2.3		Challenges	13
2.4		Summary	14
3.		**SYSTEM DESIGN**	
3.1		Problem Definition	15
3.2		Proposed Methodology	16
	3.2.1	System Architecture of Smart Walking Stick	16
3.3		Modules	18

	3.3.1	Obstacle Detection Module	18
	3.3.2	Alert System Module	18
	3.3.3	Voice Assistance Module	18
	3.3.4	Emergency button Module	18
	3.3.5	Water Detection Module	19
	3.3.6	Mobile Application Module	19
3.4		System Requirements Specifications	19
	3.4.1	Functional Requirements	19
	3.4.2	Non-Functional Requirements	20
	3.4.3	Software Requirements	21
	3.4.4	Hardware Requirements	21
3.5		Introduction to UML	22
3.6		UML Diagrams	27
4.		**SYSTEM IMPLEMENTATION**	
4.1		Technologies	32
	4.1.1	ESP32	32
	4.1.2	Ultrasonic Sensor	34
	4.1.3	Soil Moisture Sensor	35
	4.1.4	GPS Module	36
	4.1.5	MIT App Inventor	37
	4.1.6	Buzzer	39
4.2		Code	40
5.		**SYSTEM TESTING**	
5.1		Test Case Description	46
5.2		Types of Testing	49
5.3		Test Cases	52

5.4	Results	60
6.	**CONCLUSION**	**68**
7.	**FUTURE ENHANCEMENTS**	**69**
8.	**REFERENCES**	**70**

APPENDIX-A PLAGIARISM REPORT

LIST OF ABBREVIATIONS

ESP32	-	Espressif 32-bit Microcontroller
GPS	-	Global Positioning System
IoT	-	Internet of Things
TTS	-	Text-To-Speech
UML	-	Unified Modeling Language
HC-SR04	-	Ultrasonic Sensor Module
MCU	-	Microcontroller Unit
UI	-	User Interface
LED	-	Light Emitting Diode
Vcc	-	Supply Voltage
GND	-	Ground
TX	-	Transmit
RX	-	Receive
Wi-Fi	-	Wireless Fidelity
BT	-	Bluetooth
SMS	-	Short Message Service
APP	-	Application
PCB	-	Printed Circuit Board
DC	-	Direct Current
AC	-	Alternate Current
GPIO	-	General Purpose Input Output

LIST OF FIGURES

Fig. 3.2.1 System Architecture of Smart Walking Stick 17

Fig 3.6.1 Use Case Diagram..... 27

Fig 3.6.2 Class Diagram 28

Fig 3.6.3 Sequence Diagram 29

Fig 3.6.4 Data Flow Diagram..... 30

Fig 3.6.5 Activity Diagram 31

Fig 4.1.1 ESP32 Microcontroller..... 32

Fig 4.1.2 Ultrasonic Sensor 34

Fig 4.1.3 Soil Moisture Sensor..... 35

Fig 4.1.4 GPS Module..... 36

Fig 4.1.5.1 Mobile App Interface 38

Fig 4.1.5.2 App Development using MIT App Inventor..... 38

Fig 4.1.6 Buzzer Module..... 39

Fig 5.4.1 Front View of Smart Walking Stick……..60

Fig 5.4.2 Top Section with Power Switch and Emergency Button..... 60

Fig 5.4.3 Side View Showing Sensor Replacement..... 61

Fig 5.4. 4 Internal Circuit Integration of Smart Walking Stick…….61

Fig 5.4.5 Mobile Application Interface…….62

Fig 5.4.6 Front Obstacle Detection Output in Mobile Application……...62

Fig 5.4.7 Left Obstacle Detection Output in Mobile Application……63

Fig 5.4.8 Right Obstacle Detection Output in Mobile Application……..….63

Fig 5.4.9 Hole Detection Alert in Mobile Application……..64

Fig 5.4.10 Water Detection Alert in Mobile Application……..64

Fig 5.4.11 Front Obstacle Detection Output in Serial Monitor…….65

Fig 5.4.12 Left Obstacle Detection Output in Serial Monitor……...65

Fig 5.4.13 Right Obstacle Detection Output in Serial Monitor…….66

Fig 5.4.14 Hole Detection in Serial Monitor…….66

Fig 5.4.15 Water Detection in Serial Monitor……...67

Fig 5.4.16 Emergency Alert Message Sent Detection in Serial Monitor……...67

LIST OF TABLES

Table 3.4.1: Functional Requirements of Smart Walking Stick..... 19

Table 3.4.2 Non-Functional Requirements 20

Table 3.4.3 Software Requirements 21

Table 3.4.4 Hardware Requirements 21

Table 3.5.1 UML Notations 25

Table 5.3.1 Positive Test cases for Obstacle Detection Module 52

Table 5.3.2 Negative Test Cases for Obstacle Detection Module..... 53

Table 5.3.3 Positive Test Cases for Alert System Module 54

Table 5.3.4 Negative Test Cases for Alert System Module..... 55

Table 5.3.5 Positive Test Cases for Emergency Button Module 56

Table 5.3.6 Negative Test Cases for Emergency Button Module..... 57

Table 5.3.7 Positive Test Cases for Mobile Application Module 58

Table 5.3.8 Negative Test Cases for Mobile Application Module 59

CHAPTER – 1

INTRODUCTION TO SMART WALKING STICK

1.1 Overview of Smart Walking Stick

Smart Walking Stick is a smart cane for the blind, which helps them to walk and travel independently from other people. The system finds obstacles before they happen, which lowers the accident rate. Regular walking sticks only detect objects when the stick physically collides with the object. The Smart Walking Stick lets out alerts at the right time, which helps users feel more confident in previously unknown locations.

The system is controlled by a 32-bit microcontroller, an ESP32 using modern embedded systems designs, and is responsible for decoding and processing input from various sensors as well as sending out signals to other systems at the required speed and reliability.

Ultrasonic sensors are placed in several directions, such as front, left, right, and down, to find obstacles. These sensors constantly measure distances and sort obstacles into different alert levels, like medium, high, and emergency. A buzzer makes sounds to let the user know that there are obstacles nearby and how close they are.

The device also has a soil moisture sensor that can tell when there is water on a surface, which helps find wet or dangerous surfaces. A GPS module is also used to find out where the user is. In an emergency, the system can send location information to a caretaker, making sure help arrives quickly.

A mobile app made with MIT App Inventor improves the system by letting users get voice-based help through text-to-speech. It clearly tells the user which way to go around obstacles, making navigation easier. It makes it easy for the user to know where to go around obstacles. Overall, the system makes life safer, more independent, and better for people who can't see or are blind.

1.2 Need of the Project

People who are blind or have low vision have a lot of trouble getting around on their own in their daily lives. They often need help from other people, which can make them less confident and free. Traditional walking sticks are useful, but they only give limited support because they only work when they touch an obstacle. This delay in detection can make things dangerous, especially in crowded or new places.

There is a growing need for a smarter and more reliable system that can help blind people get around safely without needing help from other people all the time. A smart assistive device can help them find obstacles ahead of time and respond in the right way. This kind of system should be able to give users real-time feedback and help to make them more aware and safer.

Another serious problem with customary aids is that they do not warn the user of many hazards, such as water, holes and uneven ground, hazards that can be potentially dangerous if not detected in advance. Therefore, it is necessary to add more sensors to detect hazards to improve safety greatly.

In emergencies, people who are visually impaired may be unable to easily describe their location. This means that a system needs to be able to correctly locate visually impaired persons to alert caregivers to the fact that a person is in trouble. This can be solved with the use of GPS.The Smart Walking Stick is designed to solve all of these problems by putting together different technologies into one device. It wants to help people find their way around better, make them feel more confident, and make them less reliant on others. In general, the project meets the need for a safe, effective, and easy-to-use solution for people who are blind or have low vision.

1.3 Objectives of the Project

Our motivation is to implement a smart walking cane in order to give the blind a better sense of security and to assist them more. The system tries to warn them about the obstacles in their surrounding before contact occurs. This reduces the chance of accidents, and allows safe movement both indoors and outdoors.

Another system wish is to use several ultrasonic sensors directed in various directions and to detect obstacles. The sensors would check the space around them and determine the distance of obstacles. The information gathered by the obstacle detection system is used to give the user progressively alerts depending on how close the obstacle is, and how he should react.

The project also wants to help people find their way by using voice commands. The system tells the user exactly where obstacles are, like to the left, right, or front, by combining a mobile app with text-to-speech technology. This feature makes users more aware and makes navigation easier and more useful.

Another part of this system is a soil moisture sensor, which may be able to alert a person that there is water in the environment to prevent them from slipping or falling into water drains. There is also an emergency feature that uses the GPS module to contact a caretaker to inform them of a person in need of help.

Ideally the device would be small, inexpensive, easy to use and used on a daily basis. The aim is for the device to combine multiple functionalities into a single platform to improve independence and improve the quality of life of people who are blind or have low vision, and reduce their reliance on sighted people.

1.4 Applications of Smart Walking Stick

The Smart Walking Stick is mostly used by people who are blind or have low vision to travel independently and safely in their day-to-day activities. The Smart Walking Stick provides a safe walking experience by detecting obstacles before the person walks into them. It is most useful in situations such as walking across a room, crossing a road, or walking through a crowded public space where the user needs to avoid accidents and feel more confident while walking.

The device is very important because it makes it easier to get around outside. It can see things like walls, cars, people, and other things in the way, so the user can take the right steps to stay safe. Sound or voice guidance that tells the user which way to go also helps them be more aware of their surroundings. This makes it especially useful in places where you don't know where you're going and it's harder to find your way around.

The Smart Walking Stick can also be used in hospitals or rehabilitation centers to help people who are recovering their sight or learning to walk again to gradually get used to walking on their own. The system also alerts people when unsafe conditions such as water or uneven flooring are present.

This is really useful for emergencies too. It can also be connected to a GPS module, meaning that it can be used to send location information to family or caregivers, if the person needs to ask for help and couldn't already get to a phone. For when the person feels lost or in danger, provides a way to quickly call for help, and stay safer.

It could be used in assistive technology or the Internet of Things (IoT), where systems like this could be linked to more advanced systems that could allow blind or visually impaired people to get around a smart city or even more complicated environments. Overall, the Smart Walking Stick can be used in many different ways to make things safer and easier to get to.

1.5 Advantages of the System

There are a number of ways that the Smart Walking Stick is better than regular walking aids. One of the best things about it is that it can use ultrasonic sensors to find obstacles ahead of time. This early detection helps keep people safe and lets them act quickly. This system offers proactive safety measures, unlike regular sticks that only work when they come into contact with something.

Another advantage is that the area is sensed in front, to the left, to the right, and down, so it can sense not only obstacles in front of the user, but also traps in the ground, such as pits or manholes. With this feature the system provides a more complete safety solution.

The system becomes much more useful with the addition of voice support through a mobile app, so the user hears instructions about where the obstacles are instead of only hearing a series of buzzer tones. The use of text-to-speech technology would allow the user to interact with the system and access information.

Other features of this system include GPS emergency alerts and water leak detection. A soil moisture sensor detects wet areas of the ground, such as leaks, to reduce the likelihood of slipping or falling. The GPS emergency alert module allows the user to alert a caregiver where they are in the case of an emergency.

Last but not least , the Smart Walking Stick is meant to be affordable , portable , and simple to use .It adds a lot of features to one device without making it more complicated .This makes it easy for a lot of people to use every day .In general , the system makes life safer , more independent.

1.6 Summary

This chapter gave a thorough overview of the Smart Walking Stick system, explaining what it is and why it is important. The overview talked about how the system uses modern technology to help blind people get around safely. It also talked about the problems with traditional walking aids and why a smarter solution is needed.

The project's goals were clearly spelt out, and they were to make things safer, more independent, and better for users. The goal of the system is to find obstacles in real time, give directions, and help in case of an emergency. The project provides a complete assistive solution by combining sensors, a microcontroller, and a mobile app.

The applications of the Smart Walking Stick demonstrate its usefulness in various real-world scenarios. From daily navigation to healthcare support and emergency situations, the system plays an important role in enhancing mobility. It can also be extended to advanced assistive technologies in the future.

The Smart Walking Stick is useful in many real-life situations, as shown by its uses. The system is very important for improving mobility in everyday situations, healthcare, and emergencies. In the future, it could also be used with more advanced assistive technologies.

The system's benefits show that it works better than traditional methods. It is a reliable and easy-to-use device because it has features like multi-directional sensing, voice assistance, water detection, and GPS tracking. These benefits make visually impaired users feel safer and more confident.

To sum up, the Smart Walking Stick is a new and useful tool that helps blind people with some of their biggest problems. It uses both technology and ease of use to help people find their way around. The next chapter will be about the literature review, which talks about systems that already exist and research that has been done in this area.

CHAPTER – 2
LITERATURE REVIEW

2.1 Introduction to survey

A literature review is an important part of any research project because it helps you learn about the technologies and systems that are already in place that have to do with the project. It entails the examination of previously published research papers, articles, and technical documents to discern contemporary advancements in the field. By studying existing solutions, researchers can find out what works and what doesn't. This helps them make the proposed system better.

Several studies have been conducted on assistive technologies to support individuals who are blind or have visual impairments. To make navigation easier, a lot of systems have been made that use sensors, embedded systems, and mobile apps. The goal of these systems is to find obstacles, send alerts, and make users safer. But there are some problems with each system that need to be fixed.

The goal of these systems is to find obstacles, send alerts, and make users safer. Yet each of these systems has its own flaws that need to be addressed. The literature review helps find problems with current technologies and gives ideas for making a better system. It also helps people not do the same work twice and encourages new ideas. It is easier to choose the best methods and tools for the project when you compare different ones.

This paper reviews the literature on smart walking sticks, obstacle detection devices, and assistive devices to identify what technologies are available to help blind people move around more easily. The literature was focused mainly on systems that used a GPS module, ultrasonic sensors, and mobile applications.

The literature survey carried out for the paper is very important for making the Smart Walking Stick better and more useful overall , as it provides a good idea of the current state of research and shows the need for a better system that gets around the problems with the ones we have now .

2.2 Related Work

Related work is the research that has already been done in a certain field or area of study that is similar to the research that is being done now. A literature review or literature survey is another way to look at and analyze the previous work in a field.

Related work -1

Title: Smart Walking Stick for Visually Impaired People Using Ultrasonic Sensors and Arduino

Authors: D. Emmanuel Gbenga, A. I. Shani, A. L. Adekunle

What the Project Does

The smart walking stick works using ultrasonic sensors, which detect the object, the distance from the stick to the object, and then the buzzer beeps. This project makes it easier to walk independently for blind people. The system is used to detect obstacles earlier than waiting for the stick to make contact with the object.

Drawbacks of the Existing System

However, the system can only see obstacles on one side (the front side) , which limits its practical applications. Furthermore, it does not indicate which direction the obstacles are (left or right) . The system lacks other advanced features, such as GPS tracking, water detection, and voice-based assistance, which would further benefit navigation when used in combination.

How Our Project Overcomes These Drawbacks

Our suggested system uses several ultrasonic sensors to find obstacles from different angles. We include a mobile app that uses voice-based navigation to help users clearly understand where obstacles are. We also add GPS tracking and emergency alerts to make the app safer for users. We also added a soil moisture sensor to find water hazards, which makes our system more reliable and efficient.

2.2.1 Related work -2

Title: Development of a Smart Walking Stick for Visually Impaired People

Authors: R. Bhavani, S. Ananthakumaran

What the Project Does

This paper suggests a smart walking stick that uses the Internet of Things (IoT) to help blind people find their way safely. Ultrasonic sensors in the system find nearby obstacles, and a GPS module keeps track of the user's current location. This location information can be sent wirelessly to a caregiver in case of an emergency. Its design puts safety, remote monitoring, and basic environmental awareness at the top of the list.

Drawbacks of the Existing System

The system mainly detects obstacles in one direction, reducing its ability to provide full situational awareness. It mainly relies on buzzer alarms, which do not provide explicit information about the direction or the distance of an obstacle. Reduced user comprehension because of absence of voice guidance. The system also lacks water detection and advanced sensing, which are critical for detecting real-world hazards such as wet or slippery surfaces.

How Our Project Overcomes These Drawbacks

Our proposed system detects obstacles all around by using several ultrasonic sensors in different directions. It talks through a mobile app that makes it clear where the obstacles are and how far away, they are. Includes a sensor that measures soil moisture to find water hazards. Bluetooth communication also lets the ESP32 and the mobile app share data in real time, which makes the whole thing easier to use and more interactive.

2.2.2 Related work -3

Title: An Intelligent Walking Stick for the Blind

Authors: K. Chaitrali, D. Yogita, K. Snehal, D. Swati, D. Aarti

What the Project Does

This paper introduces a smart walking stick that provides auditory feedback to visually impaired users. The system uses sensors to find things in the way and then turns that information into sound messages. It has better interaction through audio alerts than traditional buzzer-based systems, which makes it easier and more intuitive to find your way around.

Drawbacks of the Existing System

The system doesn't work well for detecting obstacles in multiple directions, which makes it less effective in busy or complicated places. It doesn't have GPS tracking or emergency communication features, which makes it less useful in emergencies. It also doesn't have water detection or environmental sensing, which makes it less safe overall. The system might not be able to grow as much as it needs to in the future.

How Our Project Overcomes These Drawbacks

The system doesn't have advanced features like GPS tracking, emergency communication, or environmental sensing, which means it can't find holes or water. It also doesn't support wireless communication or mobile app integration, which makes it less scalable and less able to interact in real time. This work shows an intelligent walking stick that gives audio feedback to help people who can't see. Sensors in the system find the obstacles and then turn the information they find into sound messages. The mobile app has text-to-speech that gives clear voice instructions, which makes the experience better. A soil moisture sensor makes things even safer by knowing when surfaces are wet.

2.2.3 Related work – 4

Title: Smart Walking Stick for Visually Impaired

Authors: S. Mohite, M. Patel, A. Patel, V. Gaikwad

What the Project Does

This system shows a simple smart walking stick that helps blind people by using ultrasonic sensing technology. The ultrasonic sensor picks up on things in front of the user within a certain range and lets them know with a buzzer. The system is easy to use, cheap, and simple to set up, which makes it a good choice for low-cost assistive applications. It is mostly about avoiding collisions and making it easier for people to move around indoors and in simple outdoor settings.

Drawbacks of the Existing System

The system is simple but there is some problem. It only perceives impediments in one direction which makes it difficult to be aware of what is going on in the real world. Alerts from just a buzzer do not provide users enough information about the distance or direction of an obstacle, making it difficult for them to act quickly. The system does not include GPS tracking, emergency communication, and environmental sensing (e.g., water or hole detection). Moreover, it does not have wireless communication and mobile application integration, which limits its scalability and real-time interaction.

How Our Project Overcomes These Drawbacks

Our proposed system solves these problems by using multiple ultrasonic sensors to detect obstacles in all directions (front, left, right, and down), giving it a full picture of the environment. Our system doesn't just use buzzer alerts. It also uses a mobile app to send voice-based instructions through text-to-speech, which clearly shows where the obstacle is and how far away it is. We also add a soil moisture sensor to help find water, which makes things safer when it's wet. In addition, GPS tracking and an emergency alert system are built in so that carers can see where the person is at all times.

2.2.4 Related work – 5

Title: IoT-Based Smart Walking Stick with Water Detection and Safety Features

Authors: K. Gupta, V. Mehta, R. Patel

What the Project Does

The system doesn't have advanced features like GPS tracking, emergency communication, or environmental sensing, which means it can't find holes or water. It also doesn't support wireless communication or mobile app integration, which makes it less scalable and less able to interact in real time.

Drawbacks of the Existing System

The system doesn't have advanced features like GPS tracking, emergency communication, or environmental sensing, which can help you find holes or water. It also doesn't support wireless communication or integration with mobile apps, which makes it less scalable and less able to work in real time.

How Our Project Overcomes These Drawbacks

We're making this system better by adding voice help through a mobile app that clearly tells you where obstacles and bad weather are. Bluetooth lets the ESP32 and the mobile app talk to each other in real time. These changes make the system smarter, more interactive, and more useful in the real world.

2.3 Challenges

There are a lot of technical and practical problems that come up when making assistive systems for people who are blind or have low vision. Finding obstacles in all directions is one of the biggest problems. A lot of current systems only use a few sensors, which might not be able to see obstacles from all angles. This can make things unsafe, especially when there are a lot of people around or things are changing quickly.

Another big problem is giving the user clear and easy-to-understand feedback. Buzzer alerts are used by traditional systems, but they only tell you where obstacles are, not where to go. This makes it hard for people to figure out where the obstacle is. Adding voice assistance makes the system easier to use, but it needs to be set up correctly with the system.

Also, figuring out what's going on in the environment is a hard problem for these kinds of systems. You need more sensors and careful calibration to find dangers like water, holes, or uneven surfaces. Adding more sensors makes the system harder to use and uses more power, which makes it hard to keep the system running well while also making sure it is accurate.

Another important thing to think about is how to talk to each other in an emergency. In an emergency, the user should be able to quickly send their location to a job. Combining GPS modules and communication systems can be hard because they don't always connect, aren't always accurate, and need a lot of power. It's important to make sure it works well in all situations.

Finally, the cost and ease of use are two big problems that come up when making assistive devices. The system needs to be cheap, light, and easy for people who are blind to use. To make the system useful in the real world, it needs to have advanced features that are easy to use and cheap.

2.4 Summary

A comprehensive literature review of existing systems related to smart walking sticks and assistive technologies is presented in this chapter. We looked through several research papers to find out what is the latest in helping blind people navigate their way, avoid obstacles and get help. These studies provided us with valuable information on the various methods and tools applied in this field.

The related works showed what existing systems do well and what they do not do well. Some systems are better at finding obstacles, others have features like GPS tracking or voice assistance. Most systems on the other hand do not have all these features and are less useful in real life situations.

The problems talked about in this chapter show how important it is to find a better and more complete solution. To make the system work better, problems like limited sensor coverage, no directional feedback, and no emergency communication need to be fixed.

It is clear from looking at existing systems that combining different technologies can make assistive devices work much better. Combining sensors, voice assistance, and communication modules can make the experience more reliable and easier to use.

In summary, the literature review offers significant validation for the Smart Walking Stick system concept. It helps us find problems with the solutions we have now and make a system that is safer, works better, and is easier to use.

CHAPTER – 3

SYSTEM DESIGN

3.1 Problem Definition

It is very hard for people who are blind or have low vision to get around on their own in their daily lives. Most walking sticks only give physical support and only notice things when they touch them. This restriction makes it hard to see things ahead of time, which raises the risk of accidents. This is why users often need help from others, which makes them less self-sufficient. You can't see things that are in your way coming from different directions, which is another big problem. Standard aids don't show you what's on the left, right, or ground level. It's hard to get around in places where it's hard to get around, like busy streets, bumpy roads, or places you don't know. If you don't know where you're going, you could get into trouble.

In addition to finding obstacles, traditional systems don't find environmental hazards like water, pits, or open drains. These conditions can be dangerous and cause people to slip or fall. If they don't have the right tools, blind or low-vision people are still at risk of these things happening while they are moving around.

Emergencies are also very hard to deal with. Caregivers may have a hard time finding a user if they feel unsafe or need help. It's harder to get help quickly without location tracking systems, which can put you in danger.

Because of this, we need a smart and dependable system that can get around these problems. The system needs to be able to find obstacles in real time, give directions, sense the environment, and send emergency messages. These issues need to be fixed so that people who are blind or have low vision can live better, safer, and more freely.

3.2 Proposed Methodology

The proposed Smart Walking Stick combines several technologies into one device to get around the problems with current systems. An ESP32 microcontroller is used to build the system. It is the main unit that processes data from the sensors. It gets data from different sensors and gives the right outputs right away.

The system has a lot of ultrasonic sensors that look in different directions, such as up, down, left, and right. These sensors help you find things before you touch them by keeping track of how far away things are. The system puts alerts into three groups based on how far away they are: medium, high, and emergency.

A buzzer makes noise to let you know when something is close by. Along with sound alerts, a mobile app is built in to give voice-based directions. The app turns text into speech and tells the user exactly where obstacles are, which makes it easier to navigate and understand.

The system also has a sensor that can tell when the ground is wet or has water on it. This feature helps find dangerous situations and keeps people from slipping and falling. A GPS module is also built in to keep track of the user's location and help them in case of an emergency. In an emergency, the user can press a button to send their location and a message to a caretaker. This makes sure that help comes quickly and makes things safer. The proposed method combines sensing, communication, and user interaction to make an assistive system that is efficient, cheap, and easy to use.

3.2.1 System Architecture of Smart Walking Stick

The Smart Walking Stick's system architecture shows how different hardware and software parts work together to help people who can't see. The ESP32 microcontroller is the main part of the system. It gets data from sensors and processes it to make the right outputs. All the parts are connected in a way that makes sure they work smoothly and respond in real time.

There are four ultrasonic sensors in the input section: one in the front, one on the left, one on the right, and one pointing down. These sensors are always checking for obstacles and measuring how far away they are from the user. The downward sensor helps find pits or manholes, and the soil moisture sensor finds water on the ground to stop people from slipping.

The ESP32 looks at the sensor data it gets and figures out how dangerous something is based on how far away it is. The system creates different alert levels, like medium, high, and emergency, based on how close the obstacles are. The buzzer makes sounds to let the user know how close an obstacle is.

The system is also linked to a mobile app made with MIT App Inventor. The ESP32 sends data to the app over Bluetooth. The app then turns the data into voice messages using text-to-speech technology. It also has a GPS module and an emergency button that can send the user's location to a caregiver in an emergency. The system runs on a battery and a buck converter, which makes sure it works safely and reliably.

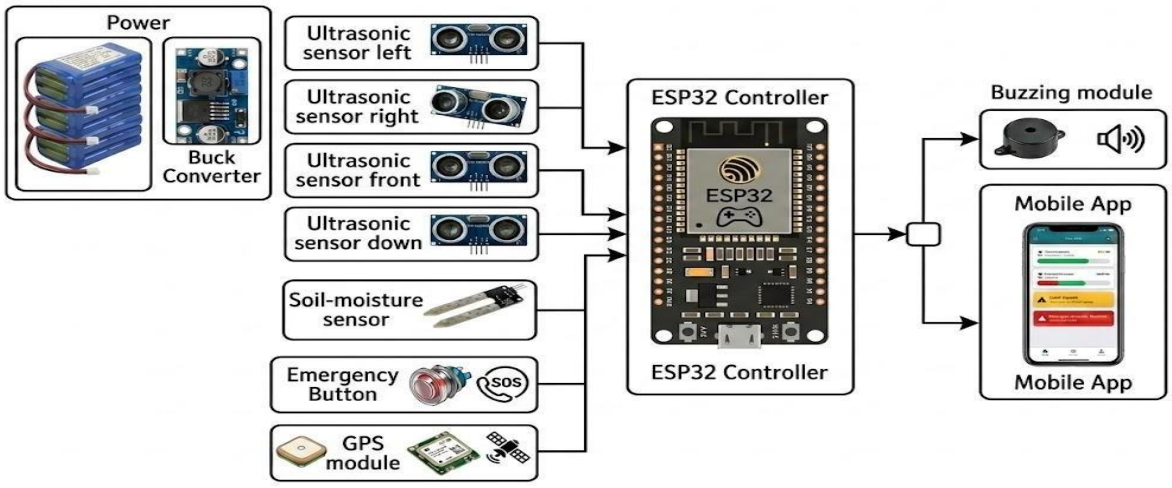


Fig 3.2.1 System Architecture

3.3 Modules

The Smart Walking Stick system is made up of several modules to make sure it works well and is easier to use. Each module does a different job, and together they help keep visually impaired users safe and give them help. Below are the main parts of the system.

3.3.1 Obstacle Detection Module

To make sure the Smart Walking Stick system works well and is better organized, it split into several modules. Each module does a different job, and together they help keep visually impaired users safe and help them. Below are the most important parts of the system.

3.3.2 Alert System Module

The alert system module sends warnings to the user based on how far away the obstacle is. The ESP32 takes the sensor data and sorts the alerts into groups like medium, high, and emergency. A buzzer makes sounds that can be different levels of loudness. This helps the user figure out how close the obstacle is and what to do about it.

3.3.3 Voice Assistance Module

This module helps users understand better by giving them voice-based instructions. It is made using a mobile app made with MIT App Inventor. The ESP32 sends data to the mobile app over Bluetooth, which then uses text-to-speech technology to turn it into speech. The system tells the user which way obstacles are, like left, right, or front.

3.3.4 Emergency Module

The emergency module keeps people safe when things get dangerous. Pressing the button sends an emergency message to a caregiver. Using a GPS module, the system finds out where the user is right now and adds that information to the message. This makes it simple for the caregiver to locate the user and assist them.

3.3.5 Water Detection Module

This module uses a soil moisture sensor to find out if there is water on the ground. Wet floors can be dangerous for people who can't see well because they can make them slip. When the system detects water, it sounds a buzzer to warn the user and help them stay away from dangerous areas.

3.3.6 Mobile Application Module

The mobile application module is a key part of making it easier for users to interact with each other. The app was made with MIT App Inventor and connects to the ESP32 over Bluetooth. It can read text out loud, which makes it easy to understand voice instructions about obstacles. It also has features for emergency communication, which makes the system smarter and easier to use.

3.4 System Requirements Specifications (SRS)

The System Requirements Specification (SRS) tells you what the Smart Walking Stick system must and must not do. It tells you what the system can do, how it works, and what it can't do. The SRS helps you understand how the system works better and makes sure that all the parts fit together well.

3.4.1 Functional Requirements

Functional requirements tell the system what it needs to do. These are features like detecting obstacles, sending alerts, voice assistance, and emergency communication.

Table 3.4.1: Functional Requirements of Smart Walking Stick

S. No	Requirement Name	Description
1	Obstacle Detection	Detect obstacles using ultrasonic sensors (front, left, right, down).
2	Distance Measurement	Measure distance between stick and obstacles continuously.
3	Alert Generation	Generate buzzer alerts based on distance levels (medium, high, emergency).
4	Direct Identification	Identify obstacle direction (left, right, front).
5	Voice Assistance	Provides voice alerts through mobile app using text-to-speech
6	Water Detection	Detect presence of water using soil moistu sensor.
7	Emergency Alert	Send emergency message with location to caretaker.

8	GPS Tracking	Provide real-time location using GPS module.
9	Bluetooth Communication	Enable communication between ESP32 and mobile app.
10	Power Management	Provide stable power using battery and buck convertor.

3.4.2 Non-Functional Requirements

Non-functional requirements tell you how well the system works, how reliable it is, and how easy it is to use.

Table 3.4.2: Non-Functional Requirements

S. No	Requirement Type	Description
1	Performance	System should respond quickly to obstacle detection (real-time processing).
2	Reliability	System should work continuously without failure.
3	Accuracy	Sensors should provide accurate distance measurements.
4	Usability	Device should be easy to use for visually impaired users.
5	Portability	System should be lightweight and easy to carry.
6	Safety	Should provide alerts to prevent accidents
7	Efficiency	Low power consumption for longer battery life.
8	Maintainability	Easy to repair and update components if needed.

3.4.2 Software Requirements

Software requirements specify the tools and technologies used to develop the system.

Table 3.4.3: Software Requirements

S. No	Software Component	Specification
1	Programming Language	C/C++ (Arduino IDE)
2	Development Platform	Arduino IDE
3	Mobile App Development	MIT App Inventor
4	Communication Protocol	Bluetooth
5	Operating System	Windows 10/11
6	Libraries Used	Ultrasonic sensor library, Bluetooth library

3.4.4 Hardware Requirements

Hardware requirements list the actual parts that make up the system

Table 3.4.4: Hardware Requirements

S. No	Hardware Component	Specification
1	Microcontroller	ESP32
2	Ultrasonic Sensors	HC-SR04 (4 sensors)
3	Soil Moisture Sensor	Water detection sensor
4	GPS Module	NEO-6M
5	Buzzer	Active buzzer
6	Battery	7.4V Li-ion battery
7	Voltage Regulator	Buck Converter (Step-down to 5V)
8	Push Button	Emergency switch
9	Connecting Wires	Wires
10	Stick Body	PVC/metal pipe

3.5 Introduction to UML (Unified Modeling Language)

Unified Modeling Language (UML) is a standardized way to use graphics to show, design, and document how a system works and what it looks like. It helps to clearly and simply show how system parts work together and how work flows. UML plays an important role in software and system development by simplifying complex designs

UML is a key part of making software and systems because it makes it easier to understand complex designs.

In the Smart Walking Stick project, UML is used to show how the sensors, ESP32 microcontroller, mobile app, and user all work together. It gives a clear picture of how the system works and helps people understand how it works better. Using UML diagrams makes it easy for developers to see how the system works and how the modules are related.

This project uses several kinds of UML diagrams, including the Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, and Data Flow Diagram. Each diagram shows a different part of the system. The Use Case Diagram shows how users interact with the system, and the Sequence Diagram shows the order in which things happen in the system.

UML also helps make system design better by cutting down on mistakes and making sure that everything is planned out before it is put into action. It gives a visual representation that makes it easier for developers, testers, and project reviewers to talk to each other. This makes for a system that is well-organized and works well.

3.5.1 Class Diagram

A Class Diagram shows how the system is set up by showing different classes, their properties, and their functions.

Classes like Mobile Application, GPS Module, Ultrasonic Sensor, and ESP32 are part of this project. It shows you how these parts fit together. This picture shows you how the system works on the inside.

3.5.2 Component Diagram

A Component Diagram shows the physical parts of a system and how they are linked together. It has both software and hardware parts. The hardware parts include sensors, an ESP32, a buzzer, and a GPS module. The software part is the mobile app. This picture shows how data goes from one part to another. It helps you see how everything fits together.

3.5.3 Deployment Diagram

A Deployment Diagram shows how the system works on real computers. It shows how the ESP32 is connected to sensors and a smartphone. It also shows how the walking stick's system is set up. This helps you understand how things really work.

3.5.4 Object Diagram

An Object Diagram shows real-time examples of parts of a system. It shows how real objects, like certain sensors or modules, work together when they are running. This picture shows what the system looked like at a certain point in time. It helps you understand how people act in real time.

3.5.5 Use Case Diagram

A Use Case Diagram shows how the user and the system work together. The user uses the stick to find obstacles and get alerts in this project. When necessary, the caretaker gets emergency messages. It helps figure out what the system can do from the user's point of view.

3.5.6 Activity Diagram

A step-by-step diagram shows how the system works. It has steps like finding things that are in the way, checking the distance, and sending alerts. It also shows you how to choose what to do based on how far away things are. This diagram helps you understand how the system works better.

3.5.7 State Machine Diagram

A State Machine Diagram shows how the system works by showing all the different states it can be in. Some states are "idle," "detecting an obstacle," "generating an alert," and "emergency mode." It tells you how the system reacts to different situations. This helps you understand how the system changes over time.

3.5.8 Sequence Diagram

A Sequence Diagram shows how different parts of a system talk to each other in the right order. It shows how sensors send data to the ESP32 and how outputs are made. It also shows you how to use the app on your phone. This picture helps you understand how to do things in the right order.

3.5.9 Collaboration Diagram

A Collaboration Diagram shows how the different parts of a system work together. It talks about how the mobile app, sensors, and ESP32 talk to each other. It shows how the parts are connected to each other. This picture will help you understand how the whole system works

Table 3.5.1: UML Notations

S.No	Symbol Name	Symbol Representation	Description
1	Class	Rectangle divided into sections	Represent a class with attributes and methods in the system
2	Actor	Stick figure	Represents the user(visually impaired person or caretaker.)
3	Use Case	Oval	Represents system functionalities like obstacle
4	Association	Line	Shows relationship between actor and the use case
5	System Boundary	Rectangle Box	Defines the boundary of the Smart Walking Stick
6	Object	Rectangle	Represents an instance such as sensor or module.
7	Message	Arrow	Represents communication between components (e.g., ESP32 to app).
8	Activity	Rounded rectangle	Represents a process like detecting obstacle or sending alert.
9	Decision	Diamond Shape	Represents condition checking (distance levels).
10	Start State	Filled Circle	Indicates start or system process.

11	End State	Circle with border	Indicates end of process.
12	Date Flow	Arrow	Represents flow of data between components.
13	Component	Rectangle with tabs	Represents hardware/software components (ESP32, sensors).
14	Node	3D box	Represents physical device like mobile or controller.
15	Control Flow	Arrow	Shows flow of execution in activity diagram.

3.6 UML Diagrams

Use Case Diagram

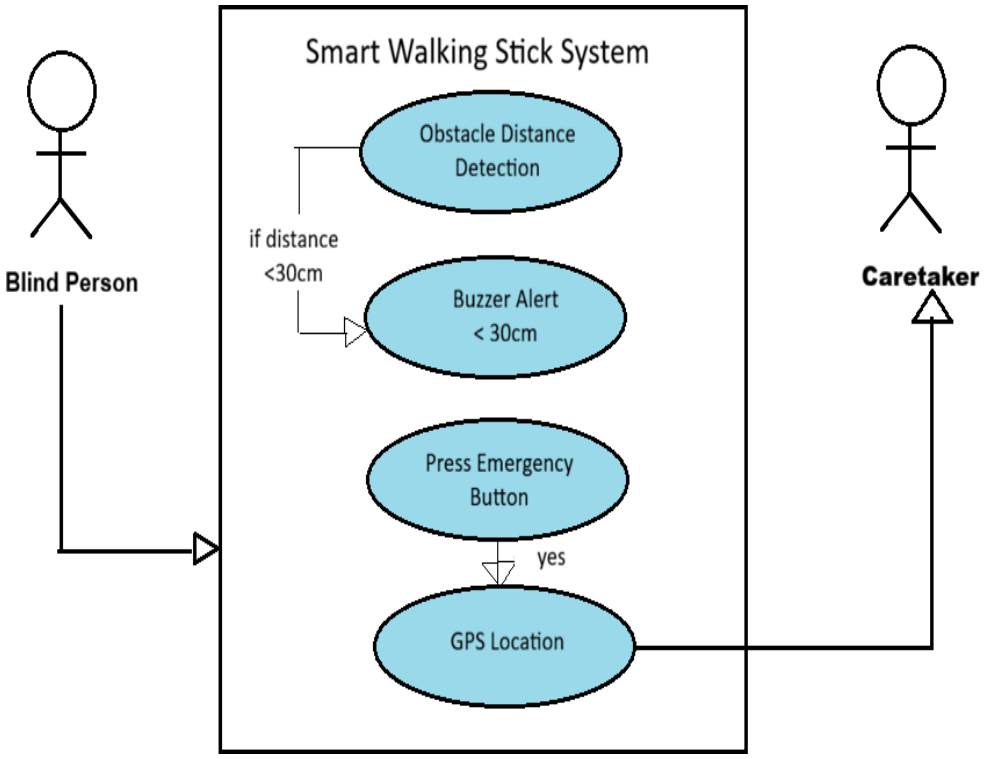


Fig 3.6.1 Use Case Diagram

The use case diagram shows how the user (a blind person) interacts with the system's features. The main character uses the smart walking stick to find things in the way and get alerts through the buzzer when things are close by. When the emergency button is pressed, the system gets the GPS location and sends it to the person who is in charge of the care. The caretaker is a second person who gets the location information in case of an emergency.

Class Diagram

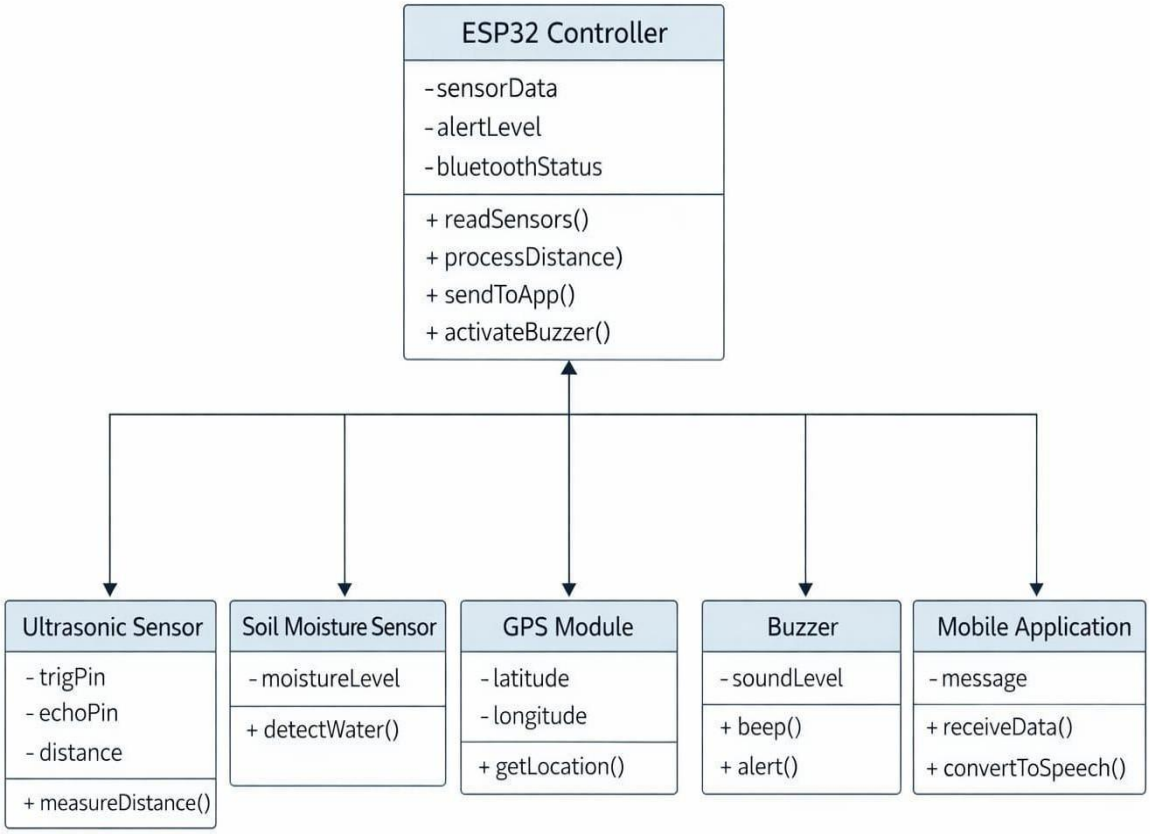


Fig 3.6.2 Class Diagram

This diagram shows clearly how users interact with the system. A class diagram shows the classes that are in a system. It displays the system's features and functions.

A class diagram shows the classes that are in a system. It displays the system's features and functions. A class diagram shows what classes are in a system. It shows what the system can do and how it works. There are four classes in the class diagram above. Each class has its own properties and methods. Three parts make up most of the class diagrams. The name of the class is Top. The second thing is the traits that make up that class. The last part is the operations of that class.

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Sequence Diagram

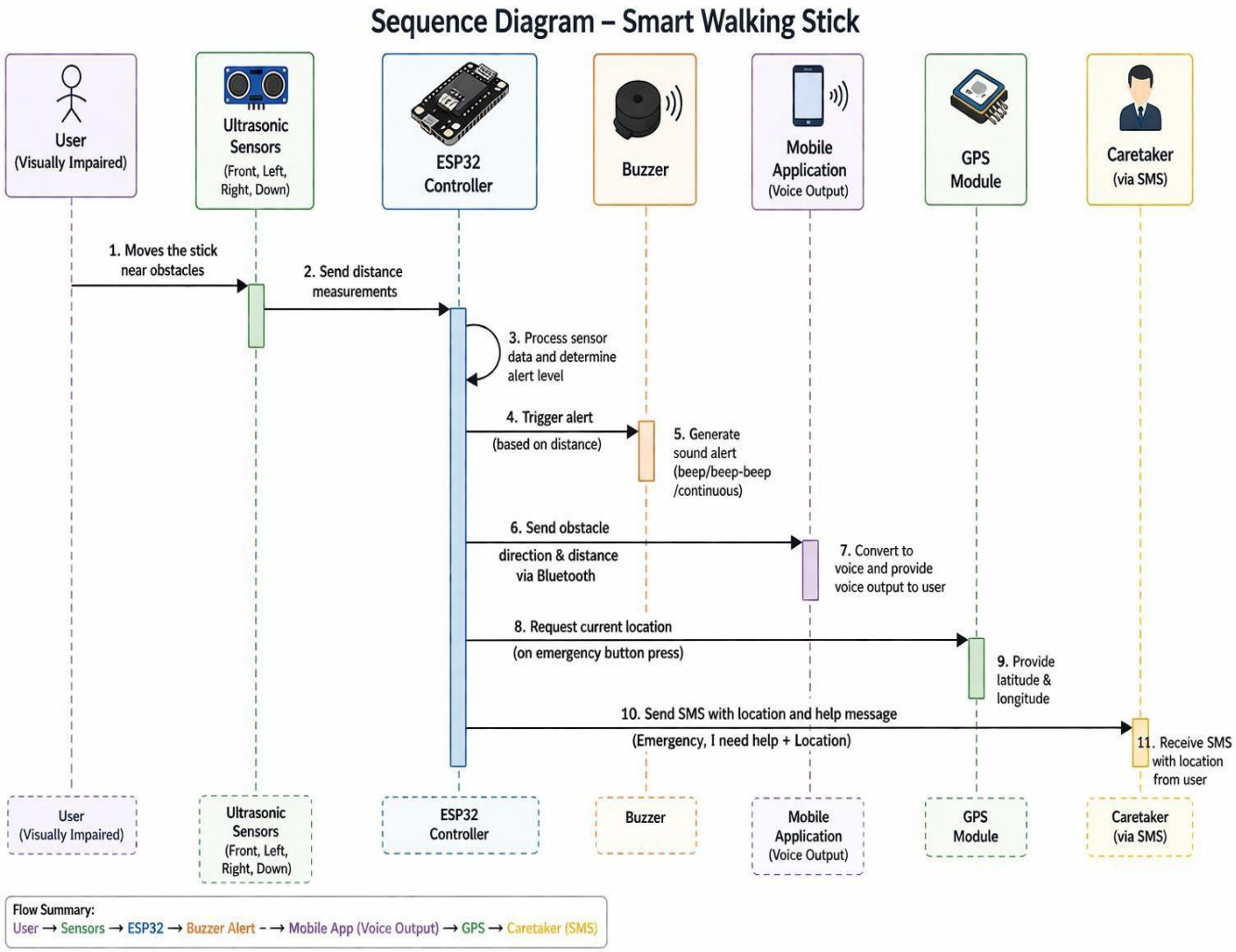


Fig 3.6.3 Sequence Diagram

A class diagram shows what classes are in a system. It shows what the system can do and how it works. There are four classes in the class diagram above. Each class has its own properties and methods. Three parts make up most of the class diagrams. The name of the class is Top. The second thing is the traits that make up that class. The last part is the operations of that class. In the architectural view model of the system being built, sequence diagrams are usually linked to use case realizations.

Data Flow Diagram

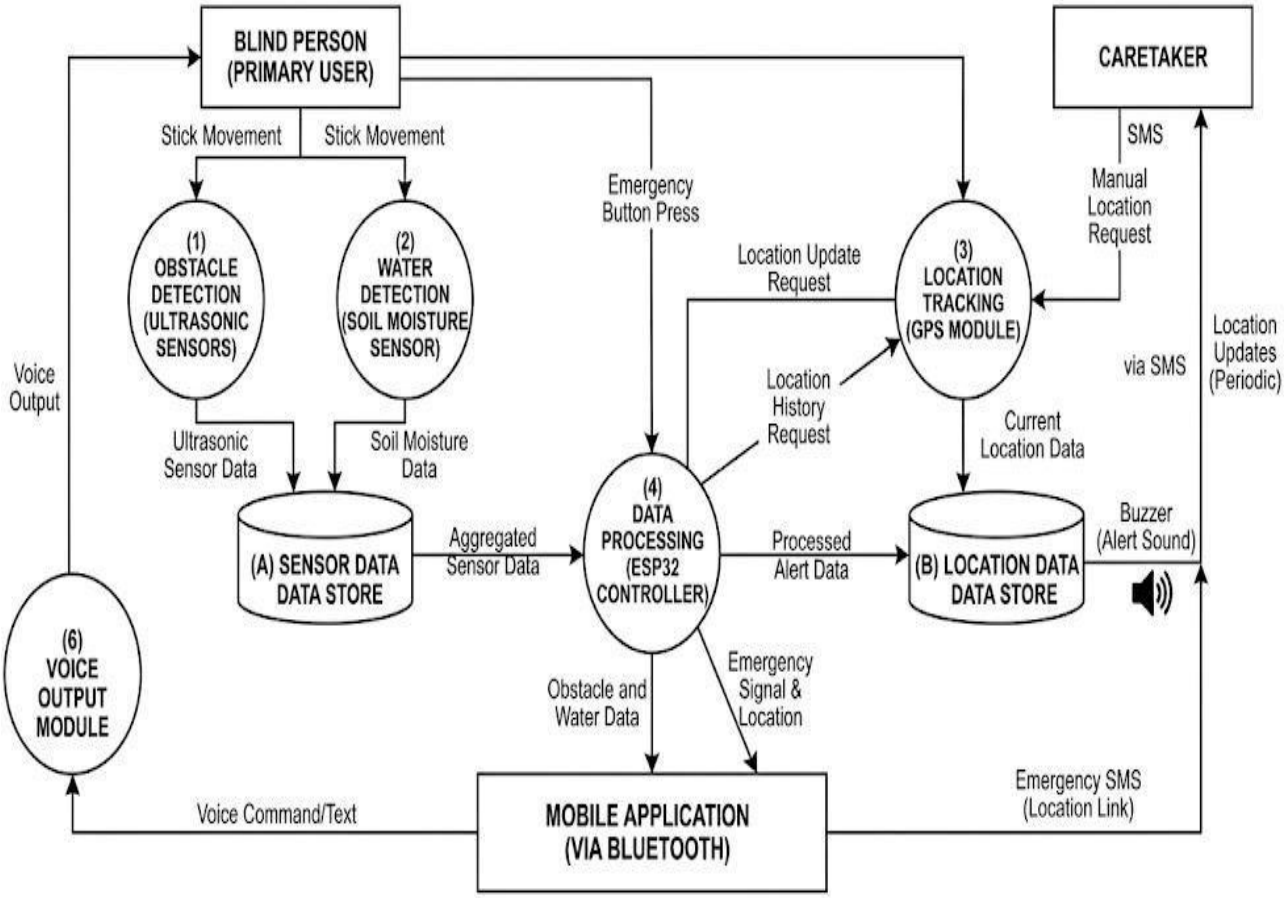


Fig 3.6.4 Data Flow Diagram

A data-flow diagram shows how data moves through a process or system, usually an information system. The DFD also shows what each entity and the process itself puts in and takes out. There are no decision rules or loops in a data-flow diagram, so there is no control flow.

Activity Diagram

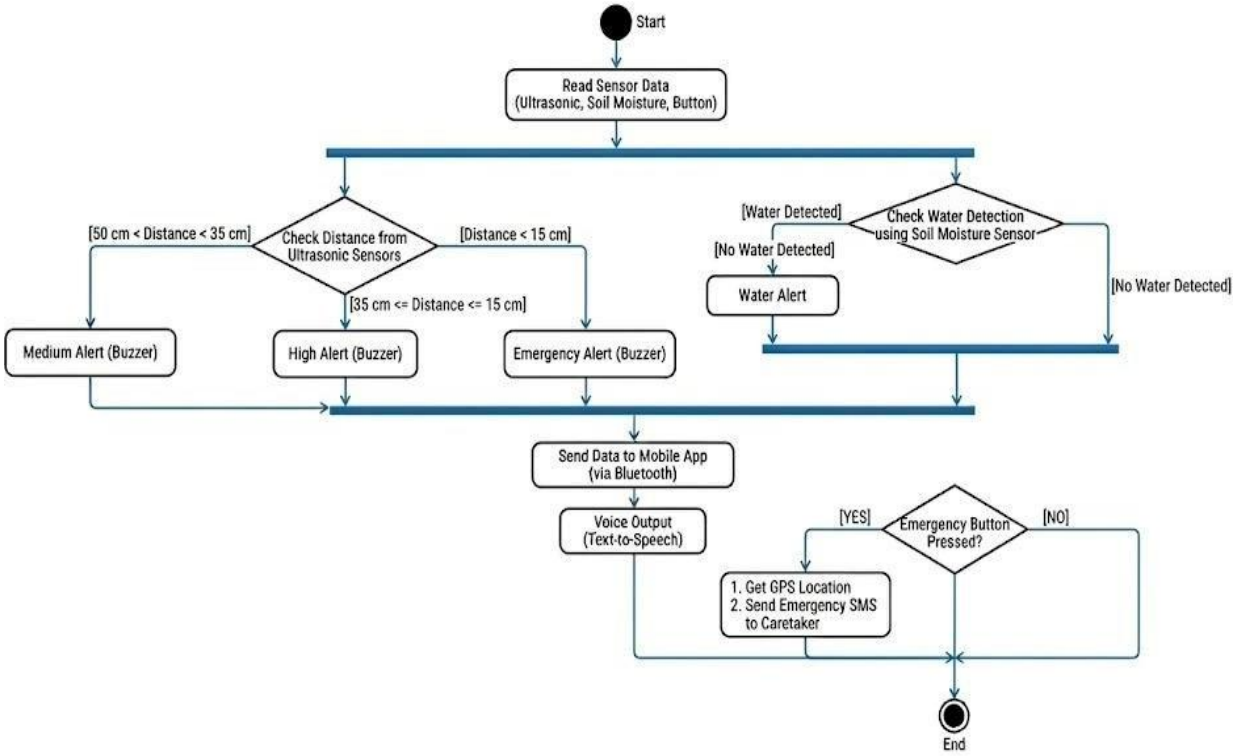


Fig 3.6.5 Activity Diagram

A flowchart is a picture that shows how a process, system, or computer algorithm works. They are used a lot in many fields to make clear, easy-to-understand diagrams of processes that are often very complicated. Flowcharts, which can also be spelled as flow charts, use rectangles, ovals, diamonds, and possibly other shapes to show what kind of step it is, and arrows to show how things should flow and in what order.

CHAPTER – 4

SYSTEM IMPLEMENTATION

4.1 Technologies

System implementation shows how the proposed smart walking stick is made with both hardware and software parts. It focuses on how each module works together to find obstacles in real time, send alerts, and communicate in an emergency. The system is meant to be easy to use, fast, and dependable for people who can't see.

Sensors, a microcontroller, communication modules, and a mobile app are the main parts of the implementation. The ESP32 is the main controller, and the ultrasonic sensors, soil moisture sensor, and GPS module send it data. The buzzer and mobile app make sounds and give voice instructions as output.

Each part is chosen because it is cheap, easy to use, and works with other parts. The integration of these technologies makes sure that the system works well in the real world. The next few sections talk about each technology used in the project.

4.1.1 ESP32

The Smart Walking Stick system's main processing unit is the ESP32 microcontroller. Espressif Systems made this powerful and efficient microcontroller, which is widely used in IoT and embedded systems. The ESP32 was chosen for this project because it can process a lot of information, has built-in Wi-Fi and Bluetooth, and has a lot of input and output pins. It is the brain of the system, making sure that all the hardware works together and works correctly.

The ESP32 gets information from a number of sensors, such as ultrasonic sensors, soil moisture sensors, and GPS modules. It uses programmed logic to make decisions based on this information in real time. For example, when an ultrasonic sensor sees something within a certain range, the ESP32 sorts the distance into different alert levels and turns on the buzzer at the right level. This processing in real time makes sure a quick response and keeps the user safer.

You can use the Arduino IDE to program the ESP32 for this project. The code is

written to keep an eye on sensor inputs and control outputs like Bluetooth communication and buzzer alerts. The ESP32 also uses Bluetooth to talk to the mobile app It sends information about the location of obstacles (front, left, right) to the app, which then turns it into voice output for the user.

One of the best things about the ESP32 is that it can replace many parts. Before, an Arduino Nano and an ESP32 were used together, but because they couldn't talk to each other, the system was made easier by only using the ESP32. This made things less complicated, more reliable, and cut down on the number of wires needed.

Also, the ESP32 needs a stable 5V power supply because it runs on a lower voltage. A buck converter is used to lower the voltage because the battery used in the project gives off 7.4V. This keeps the ESP32 and other parts safe from harm. In general, the ESP32 is very important for connecting all the modules and making the smart walking stick able to make smart decisions.

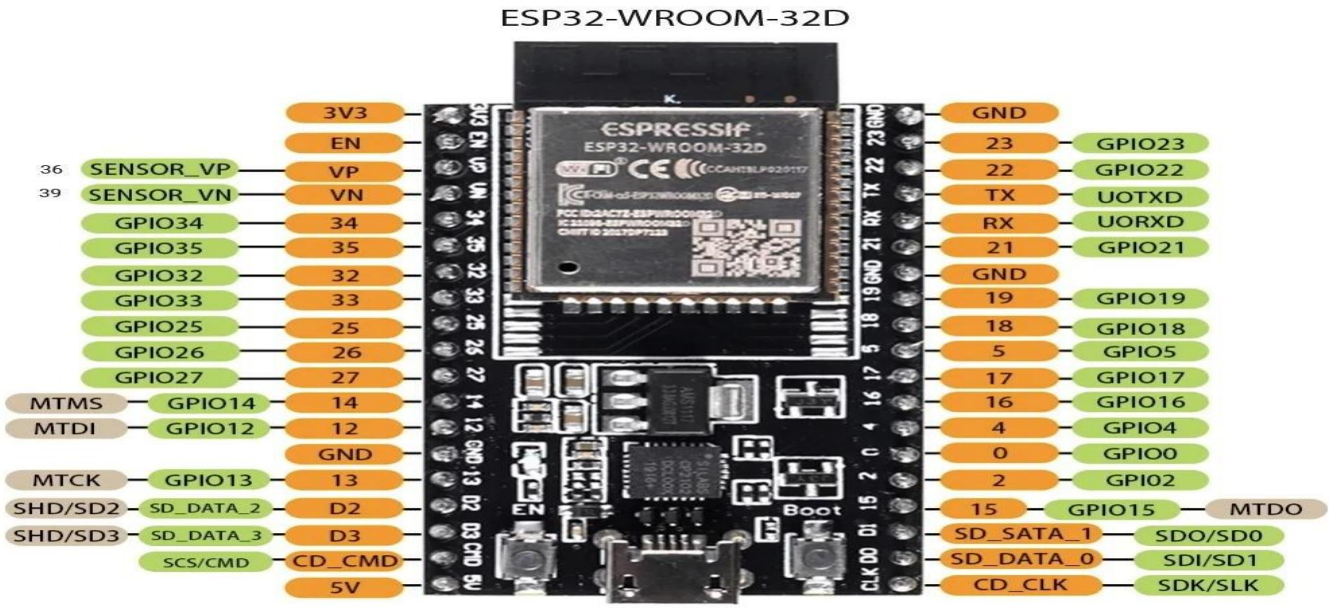


Fig 4.1. 1 ESP32 Microcontroller

4.1.2 Ultrasonic Sensor (HC-SR04)

This project uses ultrasonic sensors to find things in the way by sending sound waves through the air and measuring how far away they are. A lot of people like the HC-SR04 ultrasonic sensor because it works well, is cheap, and is easy to use. It sends and receives ultrasonic waves to find out how far away something is.

A transmitter and a receiver are the two main parts of the sensor. The transmitter sends out ultrasonic waves, which bounce off of things and come back to the receiver. A formula is used to figure out how far the waves have traveled by measuring how long it takes for them to come back. This lets the system see things without having to touch them.

The Smart Walking Stick has four ultrasonic sensors that point in different directions: forward, left, right, and down. Each sensor keeps an eye on its own area all the time. The left and right sensors see things on either side, while the front sensor sees things directly in front of them. The downward sensor is used to find pits or manholes, which is an important safety feature.

The ESP32 gets information from all four sensors and uses it to figure out how far away something is. This information is used to set up different levels of alerts. When the distance is between 35 and 50 cm, a medium alert goes off. If the distance is between 15 and 35 cm, a high alert goes off. If the distance is less than 15 cm, an emergency alert goes off.



Fig 4.1.2 Ultrasonic Sensor

4.1.3 Soil Moisture Sensor

The smart walking stick has a soil moisture sensor that can tell when there is water or wet surfaces. This feature is important because wet or slippery surfaces can be dangerous for people who can't see. The sensor helps find these kinds of situations and lets the user know ahead of time.

The soil moisture sensor works by measuring how well electricity flows through soil. When moisture touches the sensor probes, the conductivity between them goes up. The ESP32 gets an electrical signal that changes and sends it to the device. The system sends out an alert if it finds moisture.

The soil moisture sensor is put at the bottom of the walking stick in this project so that it can easily touch the ground's water. The sensor picks up on moisture when the user walks through a wet area and sends a signal to the ESP32. The system then lets the user know with a buzzer or voice output.

This feature stops people from slipping on wet floors or going into places where there is a lot of water. It's especially helpful outside, where it can be hard to see puddles of water or leaks in the drainage system.

The sensor is simple to use, inexpensive, and works well with the ESP32.It doesn't use a lot of power and gives you reliable output. Now, the system can not only find obstacles, but it can also make things safer by finding ground conditions that are dangerous.

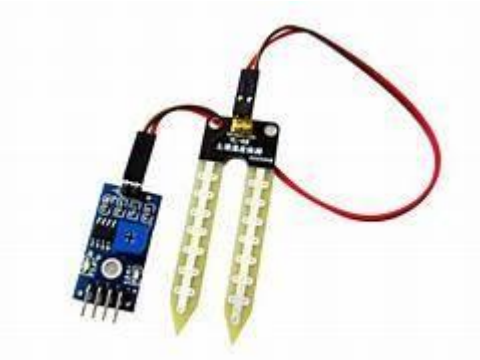


Fig 4.1.2 Soil Moisture Sensor

4.1.4 GPS Module

The project uses the GPS module to keep track of where things are in real time. It is very important in emergencies because it helps the caretaker find the user. The module usually uses satellite signals to find the device's exact location.

The GPS module gets signals from a number of satellites and uses them to figure out the current location's latitude and longitude. The ESP32 gets this information, processes it, and gets it ready to send. When needed, the location information can be sent in a message.

The GPS module is connected to the ESP32 in this project and turns on when the emergency button is pressed. The walking stick has an emergency button that the user can press if they feel unsafe or need help. When you press it, the ESP32 gets the location data from the GPS module and sends it to the caretaker.

There is a message that says "Emergency, I need help" and the sender's GPS coordinates. This makes it easier for the caregiver to find the user and help them. This feature makes things a lot safer and gets you help faster in an emergency.

The GPS module is very helpful in real life, especially when you're in a place you don't know. It makes sure that the user doesn't have to rely on others for everything and can ask for help on their own. Adding GPS to the system makes the smart walking stick much safer.

The GPS module is very helpful in real life, especially when you're in a place you don't know. It makes sure that the user doesn't have to rely on others for everything and can ask for help on their own. Adding GPS to the system makes the smart walking stick much safer.



Fig 4.1.2 GPS Module

4.1.5 MIT App Inventor

This project uses ultrasonic sensors to find things in the way by sending sound waves through the air and measuring how far away they are. A lot of people like the HC-SR04 ultrasonic sensor because it works well, is cheap, and is easy to use. It sends and receives ultrasonic waves to find out how far away something is.

The mobile app is used in this project to help the user with voice commands. The mobile app gets data from the ESP32 over Bluetooth. This data has information about the direction of the obstacle, like whether it is in front of you, to the left, or to the right.

The app turns this information into voice messages by using text-to-speech technology. If the app sees something in the way on the right side, for instance, it will say, "Obstacle on right side." This helps the user figure out exactly where the obstacle is, which is not possible with just a buzzer.

The app makes the system easier to use by giving feedback that is clear and easy to understand. It helps the user get where they need to go safely and with confidence. The interface is simple and easy to use, so anyone can use it.

One of the hard parts of the project was making the app. But MIT App Inventor made it easier to plan and build the features that were needed. The mobile app adds an intelligent communication layer to the smart walking stick, making it work better overall. The mobile app is also very important for making the system easier to use and more flexible.

It lets the ESP32 and the user talk to each other in real time over a wireless connection. Bluetooth communication uses very little power and is a safe way to send data, which makes it good for portable assistive devices like a smart walking stick.

Another important thing about the app is that it can handle more than one input at a time. It gets information from ultrasonic sensors, water detection sensors, and emergency signals, processes it, and sends out the right voice messages right away. This makes sure that the user gets information on time, which is very important for safe navigation.

In the future the app could be made better by adding features like keeping track of travel history helping with navigation or working with maps to give better directions It can also work with more than one language which makes it easier for more people to use. In the end the mobile app does more than just send voice alerts; it also connects the hardware and the user in a smart way.



Fig 4.1.5.1 Mobile App Interface

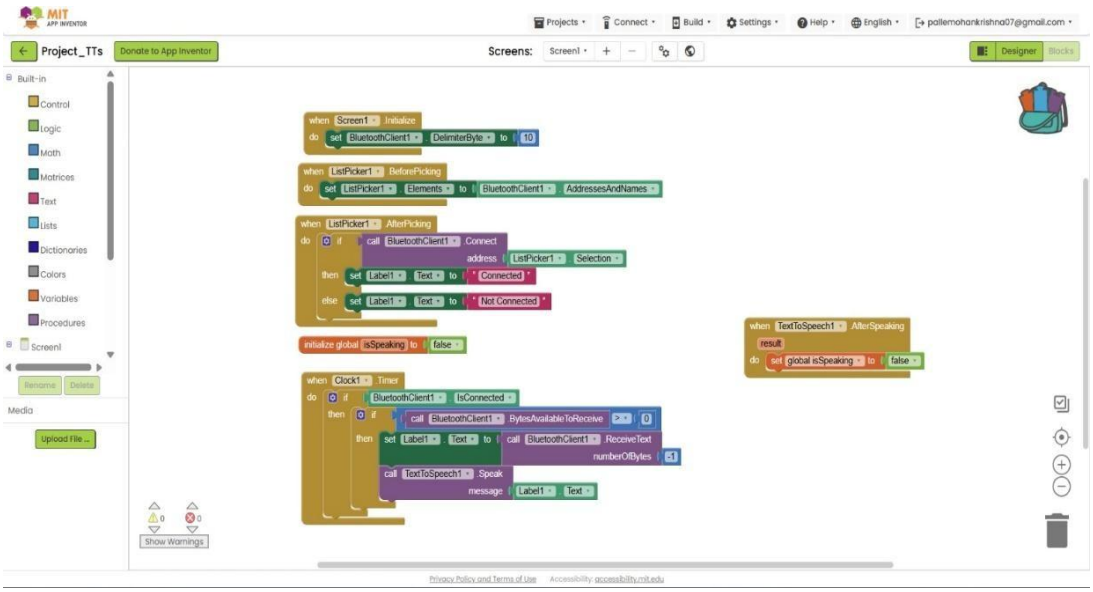


Fig 4.1.5.2 App Development using MIT App Inventor

4.1.6 Buzzer

The buzzer is an output device that makes sounds to alert the user. It is very important for letting the user know about things in the way. The buzzer is cheap, easy to use, and works well with the ESP32.The sound comes from the buzzer when it changes electrical signals.

When the ESP32 sees an obstacle with its ultrasonic sensors, it sends a signal to the buzzer. The buzzer then makes a sound that depends on how far away the object is.

This project uses different sound patterns to show how risky something is. If the obstacle isn't too far away, the buzzer makes a slow beep. As the object gets closer, the beep gets faster.

When there is an immediate threat, the buzzer makes a sound all the time. This change makes it easier for the user to see how close the obstacle is. The buzzer gives immediate feedback, which is important for real-time navigation. The system will always be able to do its job, even if you don't have the mobile app.

The buzzer works, but it doesn't tell you where the obstacle is. Adding the mobile app fixes this issue. The app and the buzzer work together to make a complete alert system. The buzzer is an important part that makes the smart walking stick work better and be easier to use.

But in places where there is a lot of noise, it can be hard to hear continuous sound alerts. This limitation shows how important it is to use the mobile app's voice and buzzer alerts together to make things clearer.



Fig 4.1.2 Buzzer

4.2 Code

```
#include <TinyGPS++.h>
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include "BluetoothSerial.h"

// -------- OBJECTS --------
BluetoothSerial SerialBT;
TinyGPSPlus gps;
HardwareSerial gpsSerial(2);

// ------ WiFi ------
const char* ssid = "YOUR_WIFI_SSID";
const char* password = "YOUR_WIFI_PASSWORD";

#define BOTtoken "YOUR_TELEGRAM_BOT_TOKEN"
#define CHAT_ID "YOUR_TELEGRAM_CHAT_ID"

WiFiClientSecure client;
UniversalTelegramBot bot(BOTtoken, client);

// ------ PINS ------
#define buttonPin 4
#define SOIL_SENSOR 34
#define buzzer 27

#define trigFront 19
#define echoFront 21
#define trigLeft 5
#define echoLeft 18
#define trigRight 25
#define echoRight 26
#define trigDown 22
#define echoDown 23

long duration;

// -------- VARIABLES --------
float latitude = 0.0, longitude = 0.0;
bool sent = false;

// -------- DISTANCE FUNCTION --------
```

```
int getDistance(int trig, int echo){
  digitalWrite(trig, LOW);
  delayMicroseconds(2);

  digitalWrite(trig, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig, LOW);

  duration = pulseIn(echo, HIGH, 20000);
  return duration * 0.034 / 2;
}

// -------- EMERGENCY FUNCTION --------
void sendEmergency() {

  Serial.println("Emergency button is pressed");

  SerialBT.end();
  delay(300);

  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(300);
  }

  client.setInsecure();

  unsigned long start = millis();
  while (millis() - start < 3000) {
  while (gpsSerial.available()) {
      gps.encode(gpsSerial.read());
    }
  }

  if (gps.location.isValid()) {
    latitude = gps.location.lat();
    longitude = gps.location.lng();
  }

  char msg[150];
  snprintf(msg, sizeof(msg),
```

```
        "EMERGENCY\nhttps://maps.google.com/?q=%.6f,%.6f",
         latitude, longitude);

  delay(500);

  bool success = bot.sendMessage(CHAT_ID, msg, "");

  if (success) {
   Serial.println("Message successfully sent to Telegram bot");
  } else {
   Serial.println("Message sending failed");
  }

  WiFi.disconnect(true);
  WiFi.mode(WIFI_OFF);

  SerialBT.begin("SMART_STICK");
}

// ------ SETUP ------
void setup() {

  Serial.begin(115200);

  gpsSerial.begin(9600, SERIAL_8N1, 16, 17);

  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(SOIL_SENSOR, INPUT);
  pinMode(buzzer, OUTPUT);

  pinMode(trigFront, OUTPUT); pinMode(echoFront, INPUT);
  pinMode(trigLeft, OUTPUT);  pinMode(echoLeft, INPUT);
  pinMode(trigRight, OUTPUT); pinMode(echoRight, INPUT);
  pinMode(trigDown, OUTPUT);  pinMode(echoDown, INPUT);

  SerialBT.begin("SMART_STICK");

  WiFi.mode(WIFI_OFF);
}

// ------ LOOP ------
void loop() {
```

```
// -------- BUTTON --------
if (digitalRead(buttonPin) == LOW && !sent) {
  sent = true;
  SerialBT.println("Emergency button is pressed");
  delay(200);
  sendEmergency();
}

if (digitalRead(buttonPin) == HIGH) {
  sent = false;
}

// -------- NORMAL MODE --------
int soil = analogRead(SOIL_SENSOR);

int front = getDistance(trigFront, echoFront);
int left  = getDistance(trigLeft, echoLeft);
int right = getDistance(trigRight, echoRight);
int down  = getDistance(trigDown, echoDown);

// ------ WATER ------
if (soil < 3500) {
  Serial.println("Water detected");
  SerialBT.println("Water detected");

  tone(buzzer, 2000);
  delay(300);
  noTone(buzzer);
}

// -------- HOLE (DOWN SENSOR) --------
if (down > 0 && down < 25) {
  Serial.println("Hole detected");
  SerialBT.println("Hole detected");

  for(int i=0; i<3; i++){
    tone(buzzer, 200);
    delay(100);
    noTone(buzzer);
    delay(100);
  }
  delay(200);
```

```
 return;
}

// -------- OBSTACLE DETECTION --------
char msg[200] = "";
int minDist = 1000;
bool first = true;

if(front > 0 && front <= 30){
  Serial.println("Front obstacle detected");

  if(!first) strcat(msg, " and ");
  sprintf(msg + strlen(msg), "Obstacle is detected at front %d cm", front);

  minDist = min(minDist, front);
  first = false;
}

if(left > 0 && left <= 30){
  Serial.println("Left obstacle detected");

  if(!first) strcat(msg, " and ");
  sprintf(msg + strlen(msg), "Obstacle is detected at left %d cm", left);

  minDist = min(minDist, left);
  first = false;
}

if(right > 0 && right <= 30){
  Serial.println("Right obstacle detected");

  if(!first) strcat(msg, " and ");
  sprintf(msg + strlen(msg), "Obstacle is detected at right %d cm", right);

  minDist = min(minDist, right);
  first = false;
}

if(strlen(msg) > 0){

  SerialBT.println(msg);
```

```
  // -------- BUZZER (ONLY FRONT, LEFT, RIGHT) --------
   if(minDist >= 20){
     tone(buzzer, 1200);
     delay(150);
     noTone(buzzer);
   }
   else if(minDist >= 10){
     tone(buzzer, 1800);
     delay(120);
     noTone(buzzer);
   }
   else{
     for(int i=0; i<3; i++){
       tone(buzzer, 2000);
       delay(80);
       noTone(buzzer);
       delay(80);
     }
   }
  }
  else {
    noTone(buzzer);
  }

  delay(50);
}
```

CHAPTER – 5

SYSTEM TESTING

Testing the system is a key part of making the Smart Walking Stick system. It makes sure that the hardware parts, like the ESP32, sensors, GPS module, and the software parts, like the mobile app, all work together. The main goal of testing is to make sure that the system works in real life and that it helps blind people safely, as it should.

Testing finds bugs, makes the system work better, and makes sure it works. To keep people safe, it's very important to test this project, so it has to be done right to make sure it doesn't break down while people are using it.

5.1 Test Case Description

The test case description explains how the Smart Walking Stick system's different features are tested in different situations. Each test case has inputs, expected outputs, and actual outputs to check if the system works as it should.

In this project, we test different parts of the system, like the GPS tracking, the emergency alert system, the mobile app communication, and the water and obstacle detection.

Purpose of Test Cases

The main objectives of test case design in this project are:

- To verify accurate obstacle detection using ultrasonic sensors
- To ensure correct water detection using soil moisture sensor
- To check buzzer alerts for different distances
- To validate Bluetooth communication between ESP32 and mobile app
- To confirm correct voice output from the mobile application
- To test emergency button functionality
- To verify GPS location tracking and SMS alert to caretaker

Test Case Structure

Each test case includes the following:

- Test Case ID
- Module Name
- Input Condition

- Expected Output
- Actual Output
- Result (Pass/Fail)

Test Case 1: Obstacle Detection

- **Input:** Object placed at distance < 30 cm
- **Expected Output:** Buzzer should produce continuous sound
- **Actual Output:** Continuous buzzer sound observed
- **Result:** Pass

Test Case 2: Medium Distance Detection

- **Input:** Object placed between 30 cm – 50 cm
- **Expected Output:** Slow beep sound
- **Actual Output:** Slow beep observed
- **Result:** Pass

Test Case 3: Water Detection

- **Input:** Soil moisture sensor touches water
- **Expected Output:** Water alert message sent to mobile app
- **Actual Output:** Voice alert "Water detected" received
- **Result:** Pass

Test Case 4: Emergency Button

- **Input:** Emergency button pressed
- **Expected Output:** GPS location sent to caretaker via SMS
- **Actual Output:** SMS received with location link
- **Result:** Pass

Test Case 5: Bluetooth Communication

- **Input:** ESP32 sends obstacle data
- **Expected Output:** Mobile app receives and converts to voice
- **Actual Output:** Voice output generated correctly
- **Result:** Pass

Test Case 6: GPS Module

- **Input:** Request for location triggered
- **Expected Output:** Latitude and longitude values retrieved
- **Actual Output:** Accurate coordinates obtained

- **Result:** Pass

Testing Environment

To make sure it worked in real time, the system was tested both inside and outside. To test the sensors' accuracy, they were placed near walls, objects, and water surfaces.

The testing environment is very important for judging how well the Smart Walking Stick system works and how reliable it is. This project tested the system in a variety of controlled and real-time situations to make sure it works correctly for visually impaired users in a variety of situations.

The hardware setup had a GPS module, a buzzer, a Bluetooth communication module, an ESP32 microcontroller, ultrasonic sensors, and a soil moisture sensor. All parts were properly connected and powered by a stable power supply to keep the tests from changing.

The software environment had Arduino IDE for programming the ESP32 and MIT App Inventor for making and testing the mobile app.

The testing environment was carefully set up to look like real life, so that the Smart Walking Stick system is safe, reliable, and useful in real life.

Conclusion of Test Case Execution

We were able to test all of the Smart Walking Stick's main features. The system worked correctly in a variety of situations, making sure that it was safe and reliable for the user.

Running test cases for the Smart Walking Stick system shows that it works well and meets its goals. All of the main features, such as detecting obstacles, detecting water, the buzzer alert system, communication with mobile apps, and emergency alerts with GPS tracking, were thoroughly tested and found to work.

In conclusion, the Smart Walking Stick system does a good job of helping people who can't see. The tests show that the system works well, is easy to use, and can handle real-world situations. When you put together sensors, a microcontroller, and a mobile app, you get a full and smart assistive solution.

5.2 Types of Testing

Different types of testing were performed to evaluate the performance and functionality of the Smart Walking Stick system.

1. Unit Testing

Unit testing is performed on individual components of the system.

In this project, each module was tested separately:

- Ultrasonic sensor tested for distance measurement
- Soil moisture sensor tested for water detection
- GPS module tested for location accuracy
- Buzzer tested for sound output
- ESP32 tested for data processing

This ensures that each component works correctly before integration.

2. Integration Testing

Integration testing verifies whether different modules work together properly.

In this project:

- Sensors were integrated with ESP32
- ESP32 was connected to mobile application via Bluetooth
- GPS module was integrated with emergency system

Testing ensured smooth communication between hardware and software components.

3. System Testing

System testing checks the complete working of the entire system.

In this phase:

- All components were tested together
- Real-time obstacle detection and alert system was verified
- Emergency alert and GPS tracking were tested simultaneously

The system was evaluated under real usage conditions.

4. Functional Testing

Functional testing ensures that each feature works as per requirements.

Examples:

- Obstacle detection triggers buzzer correctly
- Mobile app gives voice output

- Emergency button sends location

5. Performance Testing

Performance testing evaluates system efficiency and response time.

In this project:

- Sensor response time was measured
- Bluetooth communication delay was tested
- GPS response time was analysed

The system showed quick response suitable for real-time usage.

6. User Acceptance Testing (UAT)

This testing is performed to check whether the system meets user requirements.

In this project:

- The system was tested from the perspective of a visually impaired user
- Ease of use and effectiveness were evaluated
- Voice feedback and alerts were verified

The system was found user-friendly and reliable.

7. Black Box Testing

In black box testing, the system is tested without knowing internal code.

- Inputs are given and outputs are verified
- Focus is on functionality

Example:

- Input: Object placed near
- Output: Buzzer sound

8. White Box Testing

White box testing focuses on internal logic of the system.

- Code-level testing of ESP32
- Verification of conditions and loops
- Ensures correct implementation of algorithms

9. Field Testing (Real-Time Testing)

This is very important for your project.

- Tested in real environments like roads and rooms
- Checked response to real obstacles
- Verified GPS accuracy outdoors

Conclusion of Testing

All types of testing performed on the Smart Walking Stick system confirm that the system operates efficiently, reliably, and accurately under different conditions. The integration of hardware components such as ultrasonic sensors, soil moisture sensor, GPS module, and buzzer with the ESP32 microcontroller has been successfully validated through various testing methods.

The system demonstrated high accuracy in detecting obstacles at different distances and provided appropriate alert signals through the buzzer. The variation in buzzer patterns helped in distinguishing the level of danger, ensuring that the user receives timely and meaningful warnings. Similarly, the soil moisture sensor effectively identified the presence of water, which enhances user safety by preventing accidents on wet surfaces.

The communication between the ESP32 and the mobile application via Bluetooth was stable and efficient. The mobile application successfully converted sensor data into clear voice messages, allowing the user to understand the direction and nature of obstacles. This feature significantly improves navigation compared to traditional systems that rely only on sound alerts.

The emergency alert functionality proved to be highly reliable. Upon pressing the emergency button, the system was able to fetch the GPS location and send it to the caretaker through SMS without failure. This ensures that the user can receive immediate assistance in emergency situations.

In conclusion, the Smart Walking Stick system meets all the functional and performance requirements. It provides accurate detection, quick response, and reliable communication, making it a dependable assistive device for visually impaired users in real-world scenarios. The successful testing results indicate that the system is ready for implementation and can significantly improve the safety and mobility of its users.

5.3 Test Cases

Test case tables for Obstacle Detection Module

Positive Test Cases

Table 5.3.1 Positive Test Cases for Obstacle Detection Module

Function Name	Expected output	Executed output	Result status
When obstacle is detected front	Alert should be generated	Alert is generated	Successful
When obstacle is detected on left side	Alert should be generated	Alert is generated	Successful
When obstacle is detected on right side	Alert should be generated	Alert is generated	Successful
When pit or stairs are detected (down sensor)	Continuous alert should be generated	Continuous alert is generated	Successful
When multiple obstacles are detected	Alert for nearest obstacle should be generated	Nearest obstacle alert is generated	Successful
When no obstacle is present	No alert should be generated	No alert is generated	Successful

Negative Test Cases

Table 5.3.2 Negative Test Cases for Obstacle Detection Module

Function Name	Expected output	Executed output	Result status
When obstacle is present but sensor fails to detect	Alert should be generated	No alert is generated	Failed
When no obstacle is present but sensor gives false reading	No alert should be generated	Alert is generated	Failed
When obstacle is beyond detection range	No alert should be generated	No alert is generated	Failed
When sensor is disconnected or not working	System should handle without false alert	No detection / no alert	Failed
When multiple sensors fail simultaneously	Remaining sensors should work properly	No proper detection	Failed
When incorrect distance values are received due to interference	System should avoid false triggering	Incorrect alert is generated	Failed

Test case tables for Alert System Module

Positive Test Cases

Table 5.3.3 Positive Test Cases for Alert System Module

Function Name	Expected output	Executed output	Result status
When obstacle is detected by any sensor	Alert should be generated	Alert is generated	Successful
When obstacle is very close	Emergency alert should be generated	Emergency alert is generated	Successful
When obstacle is at moderate distance	High alert should be generated	High alert is generated	Successful
When pit is detected by down sensor	Emergency alert should be generated	Emergency alert is generated	Successful
When multiple obstacles are detected	Alert for nearest obstacle should be generated	Nearest obstacle alert is generated	Successful
When no obstacle is present	No alert should be generated	No alert is generated	Successful

Negative Test Cases

Table 5.3.4 Negative Test Cases for Alert System Module

Function Name	Expected output	Executed output	Result status
When obstacle is detected but alert system fails	Alert should be generated	No alert is generated	Failed
When no obstacle is present but alert is triggered	No alert should be generated	Alert is generated	Failed
When obstacle is removed	Alert should stop	Alert continues	Failed
When alert intensity does not match distance	Alert should vary based on distance	Incorrect alert pattern	Failed
When multiple sensors trigger simultaneously	Proper alert should be generated	Confusing or no alert	Failed
When there is delay in alert generation	Immediate alert should be generated	Alert is delayed	Failed

Test case tables for Emergency Button Module

Positive Test Cases

Table 5.3.5 Positive Test Cases for Emergency Button Module

Function Name	Expected output	Executed output	Result status
When user presses emergency button	Emergency alert should be triggered	Emergency alert is triggered	Successful
When GPS module is active	Correct location should be obtained	Location is obtained correctly	Successful
When emergency alert is triggered with network	Message should be sent to telegram bot	Message is sent successfully	Successful
When system is powered ON	Emergency module should be ready	Module is ready	Successful
When emergency alert is triggered	Location details should be included in message	Location is included in message	Successful

Negative Test Case

Table 5.3.6 Negative Test Cases for Emergency Button Module

Function Name	Expected output	Executed output	Result status
When emergency button is pressed but not working	Alert should be triggered	No alert is triggered	Failed
When GPS signal is unavailable	Location should not be incorrect	No location is obtained	Failed
When network is unavailable	Message should not be sent	Message not sent	Failed
When incorrect GPS data is received	Wrong location should not be sent	Incorrect location is sent	Failed
When contacts are not available	Alert should not be sent	No message is sent	Failed
When button is pressed multiple times accidentally	Multiple alerts should be controlleds	Multiple alerts are sent	Failed

Test case tables for Mobile Application Module

Positive Test Cases

Table 5.3.7 Positive Test Cases for Mobile Applicatin Module

Function Name	Expected output	Executed output	Result status
When user opens the application	Application should open successfully	Application is opened.	Successful
When Bluetooth is enabled and device selected	Device should connect successfully	Device is connected	Successful
When user gives valid voice command	Command should be executed	Command is executed	Successful
When emergency alert is received	Alert should be displayed	Alert is displayed	Successful
When GPS data is received	Location should be displayed correctly	Location displayed correctly	Successful
When continuous data is received	Data should update in real-time	Data updated continuously	Successful

Negative Test Case

Table 5.3.8 Negative Test Cases for Mobile Application Module

Function Name	Expected output	Executed output	Result status
When application fails to open	Error should be handled properly.	Application fails to open.	Failed
When Bluetooth is OFF	Connection should not be established	Connection not established	Failed
When invalid voice command is given	No action should be performed	No action is taken	Failed
When emergency alert is not received	Alert should be displayed	No alert is displayed	Failed
When GPS data is incorrect	Wrong location should not be displayed	Incorrect location displayed	Failed
When application receives unexpected data	Application should not crash	Application crashes	Failed

5.4 Results

Front View of Smart Walking Stick



Fig 5.4.1 Front View of Smart Walking Stick

Top Section with Power Switch and Emergency Button



Fig 5.4.2 Top Section with Power Switch and Emergency Button

Side View Showing Sensor Placement



Fig 5.4.3 Side View Showing Sensor Placement

Internal Circuit Integration of Smart Walking Stick

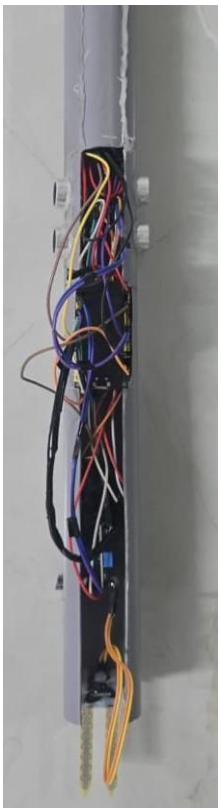


Fig 5.4.4 Internal Circuit Integration of Smart Walking Stick

Mobile Application Interface

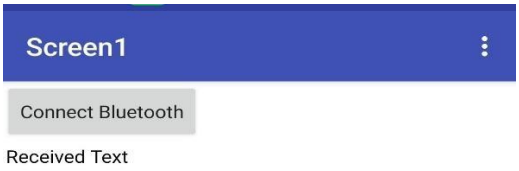


Fig 5.4.5 Mobile Application Interface

Front Obstacle Detection Output in Mobile Application



Fig 5.4.6 Front Obstacle Detection Output in Mobile Application

Left Obstacle Detection Output in Mobile Application



Fig 5.4.7 Left Obstacle Detection Output in Mobile Application

Right Obstacle Detection Output in Mobile Application

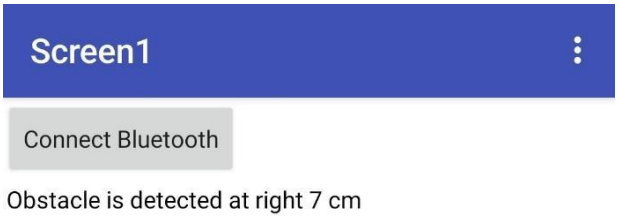


Fig 5.4.8 Right Obstacle Detection Output in Mobile Application

Hole Detection Alert in Mobile Application



Fig 5.4.9 Hole Detection Alert in Mobile

Water Detection Alert in Mobile Application

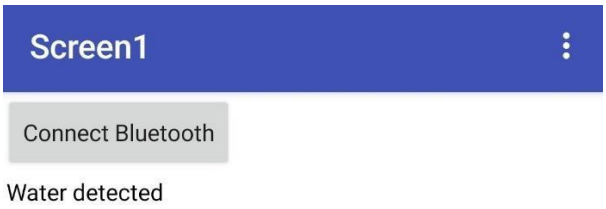


Fig 5.4.10 Water Detection Alert in Mobile Application

Front Obstacle Detection in Serial Monitor

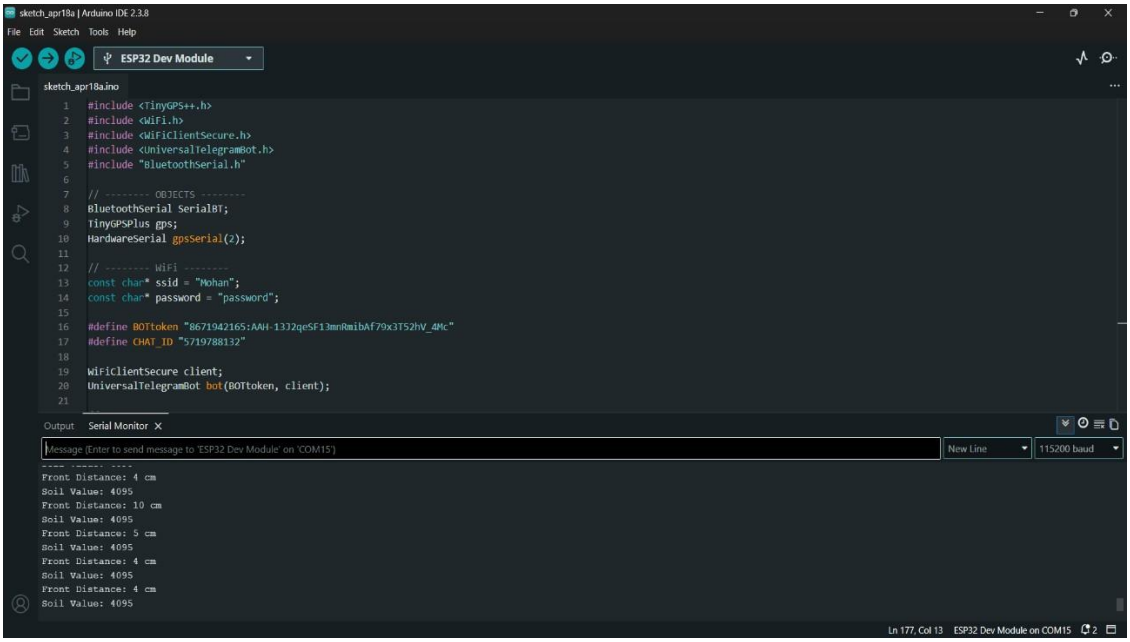


Fig 5.4.11 Front Obstacle Detection in Serial Monitor

Left Obstacle Detection in Serial Monitor

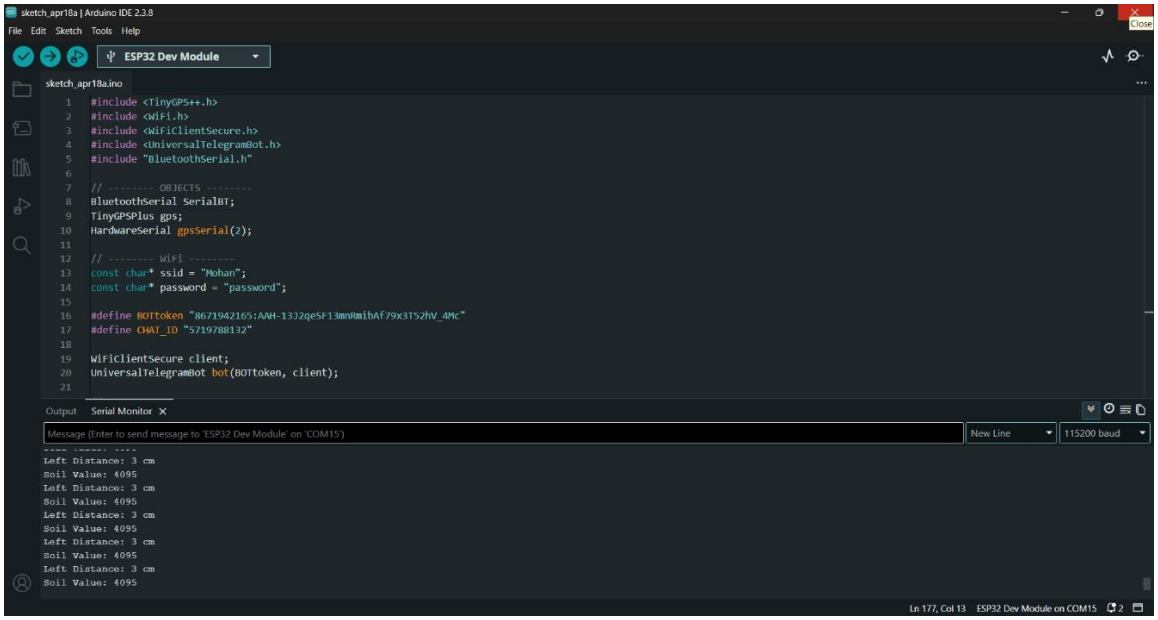


Fig 5.4.12 Left Obstacle Detection in Serial Monitor

Right Obstacle Detection in Serial Monitor

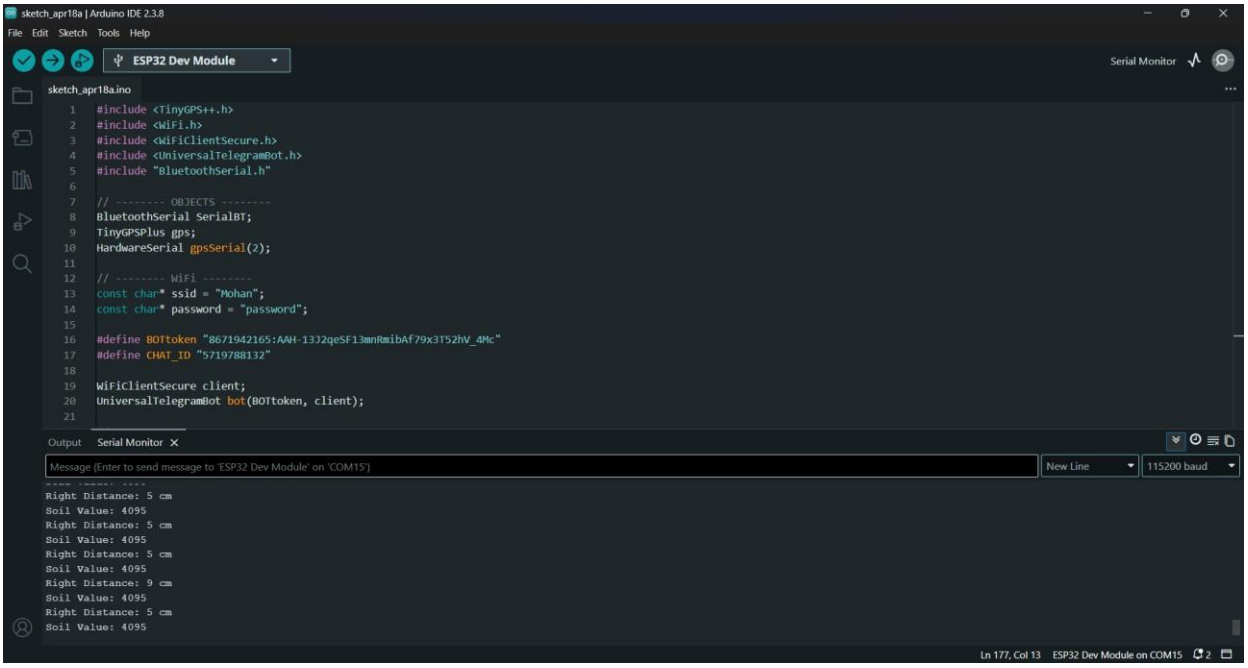


Fig 5.4.13 Right Obstacle Detection in Serial Monitor

Hole Detection Output in Serial Monitor

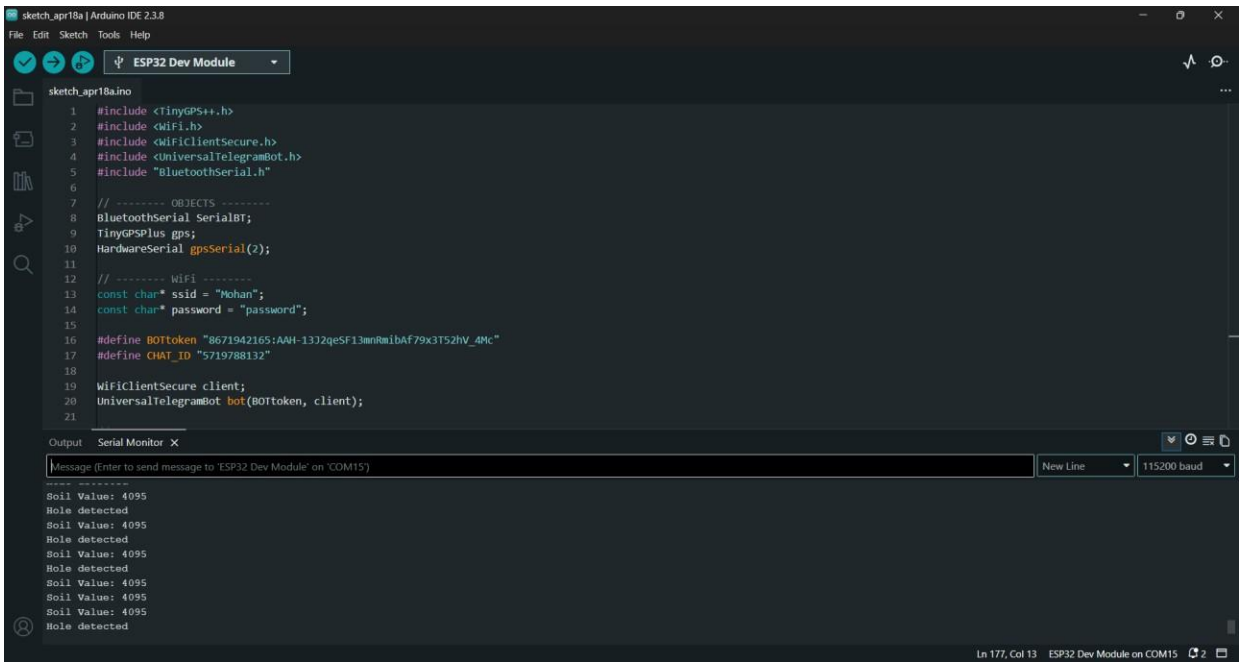


Fig 5.4.14 Hole Detection in Serial Monitor

Water Detection Output in Serial Monitor

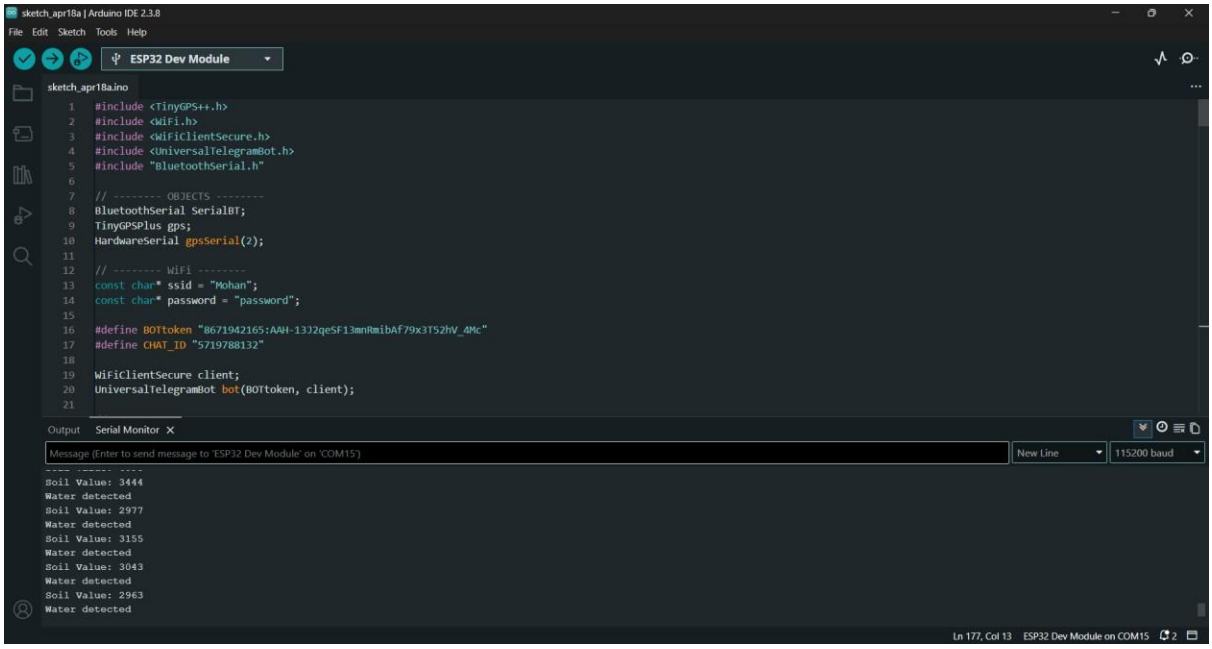


Fig 5.4.15 Water Detection in Serial Monitor

Emergency Alert Message Sent Detection in Serial Monitor

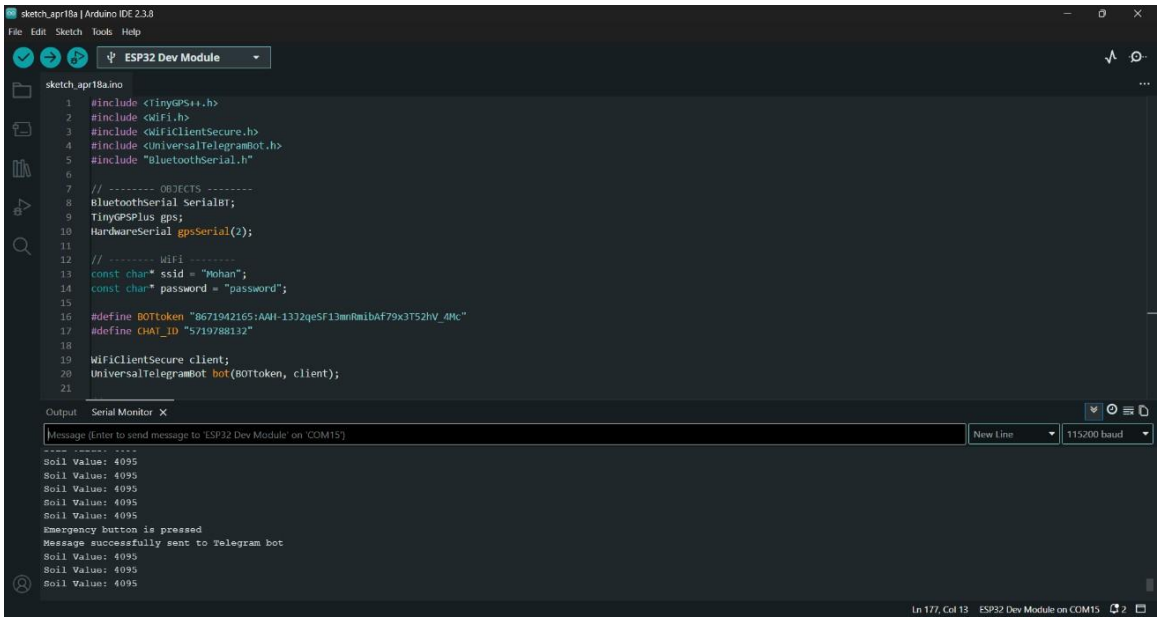


Fig 5.4.16 Emergency Alert Message Sent Detection in Serial Monitor

CHAPTER – 6
CONCLUSION

This project has successfully created a smart assistive system for people who are blind or have low vision. It has a lot of sensors and an ESP32 microcontroller. The Smart Walking Stick has a GPS module, ultrasonic sensors, a buzzer, and a mobile app to help you get where you need to go safely. It gives users up-to-the-minute information about obstacles and their surroundings, which makes them more mobile and self-sufficient. Depending on how far away they are, the ultrasonic sensors can see things in different directions and make sounds to warn you. The soil moisture sensor makes things even safer by finding wet surfaces or water that could lead to accidents.

The mobile app made with MIT App Inventor that can connect to the ESP32 via Bluetooth is one of the system's most important parts. It uses text-to-speech to give voice feedback, telling the user where things are in the way and what the weather is like. People can quickly send their GPS location to a caregiver with the emergency alert system so that help can get there quickly. The system works well and is reliable, but there are a few small problems, such as GPS lag and buzzer limits. Overall, the project is a smart, cheap, and portable solution that could be used in the real world and help improve assistive technology in the future.

CHAPTER – 7

FUTURE ENHANCEMENTS

Here are some potential areas for future enhancement of our project:

- **Smart Navigation with GPS and Maps Integration:** By combining GPS with map services like Google Maps, the system can be improved to give you directions in real time. Users can get voice directions like "turn left" or "go straight," which helps them get to certain places on
- **IoT-based Real-Time Tracking and Monitoring: :** By using IoT technology, the system can let caregivers see where the user is in real time through a mobile or web app. This makes things safer and makes it easier to keep an eye on things during emergencies.
- **Voice Command Recognition:** Voice recognition could be a feature of future systems, letting users control the device with voice commands. Users can, for example, ask for their location, check the battery level, or turn on emergency alerts without having to press buttons. This makes the system easier to use.
- **Waterproof and Rugged Design:** More sensors, like heart rate or fall detection sensors, can be added to the system. If something goes wrong or someone falls, the system can automatically send alerts to caregivers with the user's location, which makes everyone safer.

CHAPTER – 8

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